Online Appendix to "Behavioral Expectations in Nonlinear DSGE Models"

Junior Maih

Norges Bank

October 15, 2026

This Online Appendix collects the proofs, formal statements, derivations, estimation details and robustness exercises referred to in the main text, grouped by the question they answer: OA-A beliefs and foundations (the proof of Proposition 1 and the perceived-law results), OA-B risk and integrability (the proofs of Propositions 2–3 and Corollaries 1–2, and a numerical check of the risk channel), OA-C trends and regimes (the moving reference point and the proof of Proposition 4), OA-D analytical applications (the three-equation algebra of Sections 4.1 and 4.4), OA-E estimation details, OA-F lower-bound results, and OA-G the laboratory model (derivations of its equilibrium conditions, its balanced growth path, and the variants used in the paper). Each formal statement has one home: results stated in the main text are proved here under the title "Proof of ...", with the assumptions listed first; results stated only here are numbered by section (Proposition OA-A.1, ...). Equation, proposition, table and figure numbers without the prefix "OA" refer to the main text.

Contents

OA-A Beliefs and foundations OA-3
OA-A.1 Proof of Proposition 1 . . . . . OA-3
OA-A.2 The perceived-law representation OA-3
OA-A.3 Horizons OA-6
OA-A.4 Model consistency, and state versus variable contraction OA-8
OA-A.5 A sparse-max reading OA-11
OA-B Risk and integrability OA-14

OA-B.1 Proof of Proposition 2 . . . OA-14
OA-B.2 Proof of Corollary 1, and a non-Gaussian example . . . OA-14
OA-B.3 Proof of Proposition 3 . . . OA-15
OA-B.4 Proof of Corollary 2, and the threshold itself . . . OA-15
OA-B.5 Verifying the risk-discounting channel . . . OA-16
OA-C Trends and regimes . . . OA-18
OA-C.1 The moving reference point . . . OA-18
OA-C.2 Proof of Proposition 4 . . . OA-18
OA-D Analytical applications . . . OA-20
OA-D.1 Determinacy and the permanent peg . . . OA-20
OA-D.2 The forward matrix and the forward-guidance puzzle . . . OA-21
OA-E Estimation details . . . OA-23
OA-E.1 Variable-by-variable attention: details . . . OA-24
OA-E.2 The common-attention restriction: details . . . OA-26
OA-F Lower-bound results . . . OA-32
OA-F.1 The cost of the lower bound: full results . . . OA-32
OA-F.2 Attention at the lower bound: the bound as a Markov regime . . . OA-34
OA-G The laboratory model: derivations and variants . . . OA-39
OA-G.1 Agents . . . OA-39
OA-G.2 Households . . . OA-39
OA-G.3 Final-good producer . . . OA-41
OA-G.4 Firms . . . OA-41
OA-G.5 Government and monetary policy . . . OA-43
OA-G.6 Market clearing and the resource constraint . . . OA-43
OA-G.7 Balanced growth path . . . OA-43
OA-G.8 The variants at a glance . . . OA-44
OA-G.9 The detrended twin . . . OA-45
OA-G.10 The two-regime Markov bound . . . OA-47
OA-G.11 Block-specific attention . . . OA-48
OA-G.12 The term-wise rule . . . OA-49
OA-G.13 The stationary policy variant . . . OA-50

OA-A Beliefs and foundations

This section contains the proof of Proposition 1 and the formal perceived-law results summarized in Section 2.7 of the main text: the representation of the operator as a rational expectation under a perceived law (§OA-A.2), what composing it across horizons delivers (§OA-A.3), when perceived futures respect the model's relations, and how variable contraction compares with Gabaix's state contraction (§OA-A.4), and the sparse-max reading (§OA-A.5). Notation is that of Section 2.

OA-A.1 Proof of Proposition 1

Assumptions (as stated in the proposition): the behavioral equilibrium exists, is determinate, and has first-order law of motion $\hat{x}_{t+1} = A\hat{x}_t + B\varepsilon_{t+1}$, with A the transition of the behavioral equilibrium and $\hat{x}$ stacking all state variables; agents compose the one-step operator with this transition as the perceived law.

(i) For $k = 1$, (7) gives $\mathbb{E}_t^{\mathrm{BR}}\hat{x}_{t+1} = D_\rho\,\mathbb{E}_t\hat{x}_{t+1} = D_\rho A\hat{x}_t$. Suppose (8) holds at $k-1$ for forecasts made at every date. Then, because the one-step distortion is the same at every date and the operator maps a linear function of $\hat{x}_{t+1}$ to the same linear function of $D_\rho\,\mathbb{E}_t\hat{x}_{t+1}$,

$$\mathbb{E}_t^{\mathrm{BR}}\left[\mathbb{E}_{t+1}^{\mathrm{BR}}\cdots\mathbb{E}_{t+k-1}^{\mathrm{BR}}\hat{x}_{t+k}\right] = \mathbb{E}_t^{\mathrm{BR}}\left[(D_\rho A)^{k-1}\hat{x}_{t+1}\right] = (D_\rho A)^{k-1}\,\mathbb{E}_t^{\mathrm{BR}}\hat{x}_{t+1} = (D_\rho A)^k\hat{x}_t.$$

Proposition OA-A.2(ii) below gives the general form of this statement (an affine conditional mean, arbitrary higher moments) and its limits.

(ii) For $D_\rho = mI$, $(D_\rho A)^k = m^k A^k$, and $\mathbb{E}_t\hat{x}_{t+k} = A^k\hat{x}_t$ under the first-order law, so (8) is (9).

(iii) Write the rational gap and inflation equations $x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t\pi_{t+1} - r_t^n)$ and $\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa x_t$, and select x and π for behavioral treatment with common $\rho = m$. Both are level-chart variables, so by (7) every forward term $\mathbb{E}_t\{x_{t+1}, \pi_{t+1}\}$ is replaced, exactly, by $m\,\mathbb{E}_t\{x_{t+1}, \pi_{t+1}\}$, which delivers (10)–(11). In this application only x and π are discounted, so part (ii) does not apply to the agents' composed forecasts: the exogenous state r^n propagates uncontracted, and $\mathbb{E}_t^{\mathrm{BR}}\mathbb{E}_{t+1}^{\mathrm{BR}}\cdots x_{t+k} = m\,\mathbb{E}_t x_{t+k}$ (Corollary OA-A.1(b)). The horizon-k discount m^k instead arises in equilibrium by iterating (10) forward (Corollary OA-A.1(c), with the terminal condition stated there).

OA-A.2 The perceived-law representation

Setting. Fix a probability space with filtration $(\mathcal{F}_t)$; X_t, X_{t-1} and ε_t are $\mathcal{F}_t$-measurable, and $Z_t \equiv (X_t, X_{t-1}, \varepsilon_t)$ collects the arguments of f other than the forward block. Each X_i

takes values in $\mathcal{D}_i = (0,\infty)$ if $\chi_i = \log$ and $\mathcal{D}_i = \mathbb{R}$ if $\chi_i = \mathrm{id}$; $\mathcal{D} \equiv \prod_i \mathcal{D}_i$ and $\chi : \mathcal{D} \to \mathbb{R}^n$ acts componentwise. To cover regime-specific and stochastic-trend reference points, let the reference point of date $t+1$ be $\bar{X}(\xi_{t+1})$, where ξ_{t+1} is an *anchor state* (the regime r_{t+1} in Proposition 4; a stochastic trend; or a constant, in which case $\bar{X}(\xi) \equiv \bar{X}$). Attention $D_{\rho,t} = \mathrm{diag}(\rho_{X,t})$ is $\mathcal{F}_t$-measurable (constant, indexed by the current regime r_t, or a function of predetermined states, as in Remark 7). Define the *contraction map*

$$\mathcal{C}_t(\xi, y) \equiv \chi^{-1}\Big(\chi\big(\bar{X}(\xi)\big) + D_{\rho,t}\big[\chi(y) - \chi\big(\bar{X}(\xi)\big)\big]\Big), \qquad y \in \mathcal{D}, \tag{OA.1}$$

which in chart deviations is $\hat{x} \mapsto D_{\rho,t}\,\hat{x}$. It maps $\mathcal{D}$ into itself (componentwise it is $y_i \mapsto \bar{X}_i^{1-\rho_i} y_i^{\rho_i}$ or $y_i \mapsto \bar{X}_i + \rho_i(y_i - \bar{X}_i)$) and is continuous in y and in attention; it is jointly Borel when the anchor map $\xi \mapsto \bar{X}(\xi)$ is Borel, as assumed below. Mapping $\mathcal{D}$ into itself does not guarantee that the contracted vector stays in the economically meaningful domain of f (§2.2 of the main text gives a habit example); the household interpretation of the results below is therefore local. Definition 1 and (17) read

$$\mathbb{E}_t^{\mathrm{BR}} f(X_{t+1}, Z_t) \equiv \mathbb{E}_t\, f\big(\mathcal{C}_t(\xi_{t+1}, X_{t+1}), Z_t\big), \tag{OA.2}$$

with the understanding that f may also depend on ξ_{t+1} (in (17) it does, through $j = r_{t+1}$); we suppress that argument.

Definition OA-A.1 (Behavioral equilibrium; true and perceived laws)**.**

(a) *A* behavioral equilibrium *is an adapted process* (X_t) *with* $\mathbb{E}_t^{\mathrm{BR}} f(X_{t+1}, Z_t) = 0$ *almost surely for every date* t*.*

(b) *The* true law P_t *is a regular conditional distribution of* (ξ_{t+1}, X_{t+1}) *given* $\mathcal{F}_t$*, computed in that behavioral equilibrium.*

(c) *The* perceived law *is its push-forward under* $\Gamma_t(\xi, y) \equiv (\xi, \mathcal{C}_t(\xi, y))$*:* $\tilde{P}_t \equiv P_t \circ \Gamma_t^{-1}$*. Equivalently, the agent's perceived next-period state is*

$$\tilde{X}_{t+1} = \chi^{-1}\Big(\chi\big(\bar{X}(\xi_{t+1})\big) + D_{\rho,t}\big[\chi(X_{t+1}) - \chi\big(\bar{X}(\xi_{t+1})\big)\big]\Big), \tag{OA.3}$$

with the anchor state ξ_{t+1} *perceived correctly.* $\tilde{\mathbb{E}}_t$ *denotes expectation under* $\tilde{P}_t$*.*

A regular conditional distribution exists because (ξ_{t+1}, X_{t+1}) takes values in a Borel subset of a Euclidean space.

Proposition OA-A.1 (Perceived-law representation)**.** *Let* (X_t) *be a behavioral equilibrium with true law* P_t *and perceived law* $\tilde{P}_t$*.*

(i) For every jointly Borel f with $\mathbb{E}\,|f(\mathcal{C}_t(\xi_{t+1}, X_{t+1}), Z_t)| < \infty$,

$$\mathbb{E}_t^{\mathrm{BR}} f(X_{t+1}, Z_t) = \int f(y, Z_t)\, \tilde{P}_t(d\xi, dy) = \tilde{\mathbb{E}}_t f(X_{t+1}, Z_t) \qquad a.s. \tag{OA.4}$$

For $f \geq 0$ the identity holds in $[0, \infty]$ with no integrability condition; in particular $\mathbb{E}_t^{\mathrm{BR}} f$ is finite if and only if the perceived expectation is.

(ii) $\tilde{P}_t$ is the only $\mathcal{F}_t$-measurable probability kernel for which (OA.4) holds for all bounded Borel f.

(iii) (Fixed point.) (X_t) is a behavioral equilibrium if and only if it solves the equilibrium conditions (3) with $\mathbb{E}_t$ replaced by $\tilde{\mathbb{E}}_t$, where $\tilde{P}_t = P_t \circ \Gamma_t^{-1}$ and P_t is the law generated by (X_t) itself. Suppose the equilibrium is Markov, $X_t = h(X_{t-1}, \varepsilon_t)$, with $\varepsilon_{t+1} \sim \Phi$ i.i.d. and independent of $\mathcal{F}_t$, and that the anchor and attention are constant. (If attention varies, the current attention and any state that determines it are included in the Markov state, and $\mathcal{C}$ is evaluated at that attention.) Then h solves

$$\int f\Big(\mathcal{C}\big(h(h(x,e),e')\big),\, h(x,e),\, x,\, e\Big)\, \Phi(de') = 0 \qquad \text{for almost every } (x,e), \tag{OA.5}$$

almost every with respect to the law of (X_{t-1}, ε_t). The equation holds at every admissible (x, e) if equilibrium is imposed pointwise on all admissible states, or under continuity and full support. The perceived law is $\tilde{X}_{t+1} = \mathcal{C}\big(h(X_t, \varepsilon_{t+1})\big)$: the true transition of the behavioral economy*, contracted in the chart.*

(iv) At $\rho_X = \mathbf{1}$, $\mathcal{C}_t$ is the identity, $\tilde{P}_t = P_t$, and (iii) is the definition of a rational-expectations equilibrium.

Proof. (i) Append $D_{\rho,t}$ to the $\mathcal{F}_t$-measurable argument, writing $Z_t' \equiv (Z_t, D_{\rho,t})$, and let $F(\xi, y, z') \equiv f(\mathcal{C}(\xi, y; D), z)$ for $z' = (z, D)$ (suppressing the possible dependence of f on ξ). F is jointly Borel, because $\mathcal{C}$ is continuous in (y, D) and Borel in ξ (through $\bar{X}(\xi)$, assumed Borel). By the disintegration theorem for regular conditional distributions, for $\mathcal{F}_t$-measurable Z_t' and integrable (or nonnegative) $F(\xi_{t+1}, X_{t+1}, Z_t')$, $\mathbb{E}_t F(\xi_{t+1}, X_{t+1}, Z_t') = \int F(\xi, y, Z_t')\, P_t(d\xi, dy)$ a.s. By the change-of-variables formula for image measures,

$$\int f\big(\mathcal{C}_t(\xi, y), Z_t\big)\, P_t(d\xi, dy) = \int f(y', Z_t)\, \big(P_t \circ \Gamma_t^{-1}\big)(d\xi, dy').$$

Both steps hold in $[0, \infty]$ for $f \geq 0$. Combining them with (OA.2) gives (OA.4).

(ii) Let Q_t be another such kernel. Taking $f = \mathbf{1}_B$ for a Borel set B gives $Q_t(B) = \tilde{P}_t(B)$ a.s. Applying this to a countable π-system generating the Borel sets and discarding the

countable union of null sets, Q_t and $\tilde{P}_t$ agree on that π-system, hence (Dynkin) on all Borel sets, almost surely.

(iii) By (i), $\mathbb{E}_t^{\mathrm{BR}} f = \tilde{\mathbb{E}}_t f$ for every admissible f, so the two sets of equilibrium conditions coincide; $\tilde{P}_t$ is built from the law P_t of the same process. In the Markov case with constant anchor, the conditional law of X_{t+1} given $\mathcal{F}_t$ is that of $h(X_t, \varepsilon_{t+1})$ with $\varepsilon_{t+1} \sim \Phi$ independent of $\mathcal{F}_t$; substituting $X_t = h(X_{t-1}, \varepsilon_t)$ in the almost-sure equilibrium condition gives (OA.5) at $(X_{t-1}, \varepsilon_t) = (x, e)$ for almost every (x, e) under the law of (X_{t-1}, ε_t). (iv) is immediate from (OA.1). □

What is perceived, and what is not. The law that agents contract is the equilibrium law of the *behavioral* economy, not the rational-expectations law: in (OA.5) the transition h appears both inside the contraction and as the process that generates the data. This is the same fixed point as in Proposition 1, whose A is the transition of the behavioral equilibrium. Contracting the rational-expectations transition instead would describe agents who contract a law that is never realized. Two further points follow from (i)–(ii). First, the representation is exact at every order and for every f whose expectation is defined (any nonnegative or integrable f); it involves no approximation. Second, the finiteness results of Corollary 2 are statements about the perceived law: behavioral expectations are finite exactly when the perceived distribution has the relevant moment. What agents do *not* misperceive is the anchor state ξ_{t+1} (in particular the future regime), the current state Z_t, and the regime transition probabilities, which enter P_t unchanged.

OA-A.3 Horizons

The equilibrium conditions involve only one-step-ahead expectations, so Proposition OA-A.1 is all the equilibrium requires. Multi-step behavioral forecasts — the objects of Proposition 1(i) — are defined by composing the one-step operator. The next result states exactly what that composition delivers. For it, assume the Markov case of Proposition OA-A.1(iii) with a constant anchor and constant attention D_ρ (a balanced-growth anchor is handled by working with detrended variables, §OA-C.1). Write the law in chart deviations as $\hat{x}_{t+1} = \hat{h}(\hat{x}_t, \varepsilon_{t+1})$, where $\hat{x}_t$ stacks *all* state variables, including exogenous processes. Because the contraction moves the perceived state off the equilibrium support, we assume that $\hat{h}$ is a Borel transition specified on the whole domain the perceived chain reaches, off-equilibrium points included. The equilibrium law alone need not identify $\hat{h}$ there: if $z = s$ in equilibrium with $s' = as$, the transitions $(s, z) \mapsto (as, as)$ and $(s, z) \mapsto (as + c(z - s), as + c(z - s))$ generate the same equilibrium, and the same one-step beliefs under $D_\rho = \mathrm{diag}(1, m)$, but different two-step forecasts of z when $m \neq 1$ and $c \neq 0$. The results below are exact condi-

tional on the specified extension. Let $\mathsf{m}(\hat{x}) \equiv \int \hat{h}(\hat{x}, e)\, \Phi(de)$ be the true conditional mean function. For a function g of the chart deviation, define recursively $g_0 \equiv g$ and

$$g_j(\hat{x}) \equiv \int g_{j-1}\big(D_\rho\, \hat{h}(\hat{x}, e)\big)\, \Phi(de), \qquad \text{so that} \qquad \mathbb{E}_t^{\mathrm{BR}}\big[g_{j-1}(\hat{x}_{t+1})\big] = g_j(\hat{x}_t) \tag{OA.6}$$

by Proposition OA-A.1, and set $\mathbb{E}_t^{\mathrm{BR},k} g(\hat{x}_{t+k}) \equiv \mathbb{E}_t^{\mathrm{BR}} \mathbb{E}_{t+1}^{\mathrm{BR}} \cdots \mathbb{E}_{t+k-1}^{\mathrm{BR}} g(\hat{x}_{t+k}) = g_k(\hat{x}_t)$, assuming the integrals exist. Let $\Sigma_t \equiv \mathrm{Var}_t(\hat{x}_{t+1})$ and $\mu_t \equiv \mathbb{E}_t \hat{x}_{t+1} = \mathsf{m}(\hat{x}_t)$.

Proposition OA-A.2 (Horizon consistency)**.** *Under the assumptions just stated:*

(i) (Behavioral law of iterated expectations, exact.) $\mathbb{E}_t^{\mathrm{BR},k} g(\hat{x}_{t+k}) = \tilde{\mathbb{E}}_t\, g(\tilde{x}_{t+k})$, *where* $(\tilde{x}_s)_{s \geq t}$ *is the perceived chain* $\tilde{x}_{s+1} = D_\rho\, \hat{h}(\tilde{x}_s, \tilde{\varepsilon}_{s+1})$, $\tilde{x}_t = \hat{x}_t$, $\tilde{\varepsilon}_s \sim \Phi$ *i.i.d. Composing the one-step operator is therefore the same as iterating the perceived law, at every order and for every* g*; the tower property holds under the perceived measure.*

(ii) (Linear mean map, exact.) *If* $\mathsf{m}(\hat{x}) = c + A\hat{x}$ *on a set* S *with* $D_\rho \hat{h}(S, \cdot) \subset S$ *and* $\hat{x}_t \in S$*, then for every* $k \geq 1$

$$\mathbb{E}_t^{\mathrm{BR},k} \hat{x}_{t+k} = (D_\rho A)^k \hat{x}_t + \sum_{j=0}^{k-1} (D_\rho A)^j D_\rho\, c, \tag{OA.7}$$

whatever the conditional higher moments (the shocks may be non-Gaussian and their volatility state-dependent). In particular (8) *holds exactly when the conditional mean is linear with* $c = 0$ *— for instance for the first-order perturbation solution, and for any law that is affine in chart coordinates with zero intercept and mean-zero shocks, in either chart (a nonzero intercept adds the second term of* (OA.7)*).*

(iii) (Failure beyond first order.) *If* m *is quadratic,* $\mathsf{m}(y) = c + Ay + \frac{1}{2}\mathsf{M}[y, y]$ *with* M *a symmetric bilinear map (as for the conditional mean of an unpruned second-order perturbation solution), write* $\mathsf{M}[S] \equiv (\mathrm{tr}(\mathsf{M}_i S))_i$ *with* M_i *the Hessian of* m_i*. Then, exactly,*

$$\mathbb{E}_t^{\mathrm{BR},2} \hat{x}_{t+2} = D_\rho c + D_\rho A D_\rho\, \mu_t + \tfrac{1}{2} D_\rho\, \mathsf{M}\big[D_\rho \mu_t, D_\rho \mu_t\big] + \tfrac{1}{2} D_\rho\, \mathsf{M}\big[D_\rho \Sigma_t D_\rho\big]. \tag{OA.8}$$

The first-order part is $(D_\rho A)^2 \hat{x}_t$*; under the perturbation ordering (*$\hat{x}_t = O(\varsigma)$*,* $c = O(\varsigma^2)$*,* $\Sigma_t = O(\varsigma^2)$*,* M *bounded) the remaining terms are of second order and include a contracted-risk term. The curvature terms vanish for every* (μ_t, Σ_t) *if and only if* $\rho_l \rho_i \rho_j (\mathsf{M}_l)_{ij} = 0$ *for all* l, i, j*. Hence* $(D_\rho A)^k$ *is a first-order statement unless (ii) applies.*

(iv) (No horizon-invariant contraction.) *For $k \geq 2$ there is in general no single affine map, independent of the current state, that carries the family of true k-step conditional laws onto the family of perceived ones (for a fixed current state a state-dependent affine map may exist, e.g. between two Gaussian laws). In the scalar case $\hat{x}_{t+1} = a\hat{x}_t + \sigma\varepsilon_{t+1}$, $a \neq 0$, $\sigma > 0$, ε_t i.i.d. with mean zero and unit variance, $\rho \in (0,1)$, the perceived k-step mean is ρ^k times the true one, while the perceived k-step variance is $\rho^2 \sum_{i=0}^{k-1} (\rho a)^{2i}\sigma^2 > \rho^{2k} \sum_{i=0}^{k-1} a^{2i}\sigma^2$. The cumulant scaling of Proposition 3 is a one-step-ahead property.*

Proof of Proposition OA-A.2. (i) Let $\tilde{P}(\hat{x}, B) \equiv \Phi\{e : D_\rho \hat{h}(\hat{x}, e) \in B\}$ be the perceived transition kernel; by change of variables, $(\tilde{P}g)(\hat{x}) = \int g(D_\rho \hat{h}(\hat{x}, e))\, \Phi(de)$, so (OA.6) reads $g_j = \tilde{P} g_{j-1}$ and $g_k = \tilde{P}^k g$. Since $\tilde{P}^k$ is the k-step transition kernel of the perceived chain, $g_k(\hat{x}_t) = \tilde{\mathbb{E}}_t g(\tilde{x}_{t+k})$.

(ii) Let $s_k \equiv \sum_{j=0}^{k-1} (D_\rho A)^j D_\rho c$. We show by induction that $g_k(\hat{x}) = (D_\rho A)^k \hat{x} + s_k$ on S for $g = \mathrm{id}$. For $k = 1$, $g_1(\hat{x}) = D_\rho \mathsf{m}(\hat{x}) = D_\rho A\hat{x} + D_\rho c$. If the claim holds at $k-1$, then, since $D_\rho \hat{h}(\hat{x}, e) \in S$, $g_k(\hat{x}) = \int \left[(D_\rho A)^{k-1} D_\rho \hat{h}(\hat{x}, e) + s_{k-1}\right] \Phi(de) = (D_\rho A)^{k-1} D_\rho (c + A\hat{x}) + s_{k-1} = (D_\rho A)^k \hat{x} + s_k$. Only the conditional mean of $\hat{h}$ enters. A first-order perturbation solution has $\hat{h}(\hat{x}, e) = A\hat{x} + Be$, so $c = 0$ and $S = \mathbb{R}^n$.

(iii) By (OA.6), $\mathbb{E}_t^{\mathrm{BR},2} \hat{x}_{t+2} = \mathbb{E}_t^{\mathrm{BR}}[D_\rho \mathsf{m}(\hat{x}_{t+1})] = D_\rho\, \mathbb{E}_t \mathsf{m}(D_\rho \hat{x}_{t+1})$. With $Y \equiv D_\rho \hat{x}_{t+1}$, $\mathbb{E}_t Y = D_\rho \mu_t$, $\mathrm{Var}_t Y = D_\rho \Sigma_t D_\rho$ and, for quadratic m, $\mathbb{E}_t \mathsf{m}_i(Y) = \mathsf{m}_i(\mathbb{E}_t Y) + \frac{1}{2}\mathrm{tr}(\mathsf{M}_i \mathrm{Var}_t Y)$ exactly; this gives (OA.8). Under the perturbation ordering, c and M terms are $O(\varsigma^2)$ and $\mu_t = A\hat{x}_t + O(\varsigma^2)$, so the first-order part is $D_\rho A D_\rho A \hat{x}_t$. The curvature terms equal $\frac{1}{2} D_\rho \mathsf{M}[D_\rho(\mu_t \mu_t^\top + \Sigma_t) D_\rho]$; as $\mu_t \mu_t^\top + \Sigma_t$ ranges over the positive semidefinite matrices, which span the symmetric matrices, they vanish identically if and only if $\rho_l D_\rho \mathsf{M}_l D_\rho = 0$ for every l, i.e. $\rho_l \rho_i \rho_j (\mathsf{M}_l)_{ij} = 0$.

(iv) The perceived chain is $\tilde{x}_{t+k} = (\rho a)^k \hat{x}_t + \sum_{j=1}^{k} (\rho a)^{k-j} \rho\sigma\tilde{\varepsilon}_{t+j}$, the true one $\hat{x}_{t+k} = a^k \hat{x}_t + \sum_{j=1}^{k} a^{k-j}\sigma\varepsilon_{t+j}$. A single affine map $y \mapsto \lambda y + d$, independent of $\hat{x}_t$, carrying every true k-step conditional law to the perceived one must match means for all $\hat{x}_t$, so $\lambda = \rho^k$, $d = 0$, and then must scale the variance by ρ^{2k}. (For a fixed $\hat{x}_t$ a state-dependent shift would be available; the claim concerns the family of laws.) With $\sigma > 0$ and i.i.d. mean-zero, unit-variance innovations, $\rho^2 \sum_{i<k} (\rho a)^{2i} - \rho^{2k} \sum_{i<k} a^{2i} = \sum_{i<k} a^{2i} \rho^{2i+2} \left(1 - \rho^{2(k-1-i)}\right) > 0$ for $k \geq 2$, since the $i = 0$ term is strictly positive. □

OA-A.4 Model consistency, and state versus variable contraction

Under the perceived law each forward variable is contracted by its own attention. If the equilibrium ties one variable to others — a jump variable is a function of the states, or an identity links several variables — the perceived vector need not satisfy that relation.

Proposition OA-A.3 (Model consistency of the perceived law)**.** *Suppose that in the behavioral equilibrium $\hat{x}_{i,t+1} = \psi(\hat{x}_{J,t+1})$ P_t-a.s. for a Borel ψ and an index set $J \not\ni i$. The perceived law satisfies the same relation, $\tilde{x}_{i,t+1} = \psi(\tilde{x}_{J,t+1})$ $\tilde{P}_t$-a.s., if and only if $\rho_i\, \psi(y) = \psi(D_{\rho,J}\, y)$ for P_t-almost every value y of $\hat{x}_{J,t+1}$. In particular:*

(a) If $\psi(y) = b^\top y$ is linear, the relation is respected whenever $\rho_j = \rho_i$ for every $j \in J$ with $b_j \neq 0$; if the support of $\hat{x}_{J,t+1}$ spans $\mathbb{R}^{|J|}$, this condition is also necessary.

(b) If attention is common, $\rho_i = \rho_j = \rho \in (0,1)$ for $j \in J$, ψ is differentiable at 0, and the relation must hold for all y in an open ball B around 0 (for instance, $\hat{x}_{J,t+1}$ has a positive density on B and ψ is continuous), then it is respected only if ψ is linear on B.

Proof of Proposition OA-A.3. Under $\tilde{P}_t$, $(\tilde{x}_i, \tilde{x}_J) = (\rho_i \hat{x}_i, D_{\rho,J}\hat{x}_J) = (\rho_i \psi(\hat{x}_J), D_{\rho,J}\hat{x}_J)$, which satisfies $\tilde{x}_i = \psi(\tilde{x}_J)$ iff $\rho_i \psi(\hat{x}_J) = \psi(D_{\rho,J}\hat{x}_J)$. (a) The condition reads $\sum_{j\in J} b_j(\rho_i - \rho_j)y_j = 0$ on the support, which holds if every coefficient vanishes and, when the support spans, only then. (b) Setting $y = 0$ gives $\psi(0) = \rho\psi(0)$, so $\psi(0) = 0$. For $y \in B$, $\rho^k y \in B$ and induction on $\psi(\rho z) = \rho\psi(z)$ gives $\psi(\rho^k y) = \rho^k \psi(y)$. Differentiability at 0 gives $\psi(\rho^k y) = \rho^k \nabla\psi(0)^\top y + o(\rho^k)$; dividing by ρ^k and letting $k \to \infty$, $\psi(y) = \nabla\psi(0)^\top y$. □

Proposition OA-A.3 locates precisely where the operator departs from Gabaix's construction. Gabaix (2020) contracts the *state*: in deviations from the steady state, the true transition $s_{t+1} = A_s s_t + B\varepsilon_{t+1}$ is perceived as $\tilde{s}_{t+1} = m(A_s s_t + B\varepsilon_{t+1})$, and endogenous variables are the equilibrium functions of the perceived state, so perceived futures are model-consistent by construction. The operator contracts the forward *variables* themselves. With common attention m on the relevant jump variables ($D_z = mI$), the two constructions agree on one-step beliefs at first order; composed forecasts can differ unless the state contraction is matched as well ($D_s = mI$, part (b)). Beyond first order the constructions can differ; with nonzero quadratic curvature of the equilibrium maps they generally differ already at second order. Nonlinearity alone does not force a second-order difference: for $\psi(s) = s + s^3$, variable contraction gives $ms + ms^3$ and state contraction $ms + m^3 s^3$, which agree through second order and differ at third order for $0 < m < 1$.

Corollary OA-A.1 (State versus variable contraction)**.** *Let the first-order behavioral equilibrium be $s_{t+1} = A_s s_t + B\varepsilon_{t+1}$, $z_t = \Phi s_t$, with states s (exogenous and predetermined) and jumps z, $\hat{x} = (s, z)$ and $D_\rho = \mathrm{diag}(D_s, D_z)$; off the equilibrium relation $z = \Phi s$ the transition is extended by the convention that current jumps do not affect it, $\hat{x}_{t+1} = A\hat{x}_t + (B; \Phi B)\varepsilon_{t+1}$ with $A = \left(\begin{smallmatrix} A_s & 0 \\ \Phi A_s & 0 \end{smallmatrix}\right)$ (a choice, not something the equilibrium law identifies).*

(a) *If $D_z = mI$, the operator's one-step perceived law of z_{t+1} is $m\Phi(A_s s_t + B\varepsilon_{t+1})$, identical to Gabaix's, whatever D_s. Equilibrium conditions that involve only one-step expectations of z are therefore the same under both constructions.*

(b) *The composed forecasts are $\mathbb{E}_t^{\mathrm{BR},k} z_{t+k} = D_z \Phi A_s (D_s A_s)^{k-1} s_t$. With $D_z = mI$ and $D_s = I$ this is $m\,\mathbb{E}_t z_{t+k}$ for every $k \geq 1$: the attenuation does* not *compound. It equals Gabaix's $m^k \mathbb{E}_t z_{t+k}$ if $D_s = mI$.*

(c) *In the three-equation model with x, π selected at common $m < 1$ and a solution with bounded conditional means (so that $m^K \mathbb{E}_t x_{t+K} \to 0$; at $m = 1$ this terminal condition must be imposed), iterating (10) forward under the rational law of iterated expectations gives*

$$x_t = -\sigma \sum_{k \geq 0} m^k \, \mathbb{E}_t \big[i_{t+k} - m\, \pi_{t+k+1} - r^n_{t+k} \big], \tag{OA.9}$$

so the equilibrium carries Gabaix's horizon-k discount m^k through the one-step conditions, independently of D_s.

Proof of Corollary OA-A.1. (a) The operator perceives $z_{t+1} = \Phi s_{t+1}$ as $D_z \Phi(A_s s_t + B\varepsilon_{t+1})$; Gabaix's agent perceives $\Phi \tilde{s}_{t+1} = m\Phi(A_s s_t + B\varepsilon_{t+1})$. (b) In the stacked form $\hat{x}_{t+1} = A\hat{x}_t + (B; \Phi B)\varepsilon_{t+1}$, by induction on k,

$$A = \begin{pmatrix} A_s & 0 \\ \Phi A_s & 0 \end{pmatrix}, \qquad D_\rho A = \begin{pmatrix} D_s A_s & 0 \\ D_z \Phi A_s & 0 \end{pmatrix}, \qquad (D_\rho A)^k = \begin{pmatrix} (D_s A_s)^k & 0 \\ D_z \Phi A_s (D_s A_s)^{k-1} & 0 \end{pmatrix};$$

apply Proposition OA-A.2(ii) with $c = 0$. With $D_s = I$, $D_z = mI$ the z-block is $m\Phi A_s^k s_t = m\mathbb{E}_t z_{t+k}$; with $D_s = D_z = mI$ it is $m^k \Phi A_s^k s_t$. (c) By (7), $\mathbb{E}_t^{\mathrm{BR}} x_{t+1} = m\mathbb{E}_t x_{t+1}$ and $\mathbb{E}_t^{\mathrm{BR}} \pi_{t+1} = m\mathbb{E}_t \pi_{t+1}$ (exactly: x, π are level-chart variables). Substituting (10) at $t+1, \ldots, t+K-1$ and using $\mathbb{E}_t \mathbb{E}_{t+j} = \mathbb{E}_t$ yields

$$x_t = -\sigma \sum_{k=0}^{K-1} m^k \mathbb{E}_t \big[\, i_{t+k} - m\pi_{t+k+1} - r^n_{t+k} \big] + m^K \mathbb{E}_t x_{t+K},$$

and the remainder vanishes as $K \to \infty$ when $m < 1$ and the conditional means $\mathbb{E}_t x_{t+K}$ are bounded. At $m = 1$ boundedness is not enough — a bounded nonzero terminal term can survive — and $\lim_{K\to\infty} \mathbb{E}_t x_{t+K} = 0$, with convergence of the sum, must be imposed. □

The practical content of (b) is that the "common discount" $\rho_X = m\mathbf{1}$ in Proposition 1(ii) must include the exogenous states for composed forecasts to reproduce m^k. But in the three-equation application exogenous states never appear at $t+1$ in the equilibrium conditions,

so their attention is not pinned down by that model or identified by its likelihood. The robust statement of the nesting is (a) together with (c): the operator reproduces Gabaix's one-step beliefs and, through the forward solution of its one-step conditions, his m^k horizon discounting.

OA-A.5 A sparse-max reading

Gabaix's (2014) sparse max (extended to dynamic problems in Gabaix, 2016a) proceeds in two steps: the agent (1) chooses attention $m \in [0,1]^n$ by minimizing a quadratic approximation of the loss from inattention plus a cost of attention, and then (2) maximizes *exactly* under the perceived variables $m_i x_i + (1-m_i)x_i^d$, where x^d is a default. Take the default to be the reference point and apply the perception rule to the realization of the forward variable. The next result shows that step (2) is the operator exactly, and step (1) delivers a closed form for ρ, but only under stated approximations.

Proposition OA-A.4 (Sparse-max reading)**.** *An agent chooses an $\mathcal{F}_t$-measurable action a in an open set of $\mathbb{R}^p$ to maximize $\mathbb{E}_t u(a, Y_{t+1})$, where $Y_{t+1} \in \mathbb{R}^n$ are chart deviations of variables whose law the agent takes as given, and u is C^3 with bounded third derivatives. The first-order condition $\mathbb{E}_t \nabla_a u(a, Y_{t+1}) = 0$ is an equilibrium condition of the form (3). Perceptions are $Y^s = D_m Y$ (default $Y^d = 0$, i.e. the reference point), and $a(m)$ maximizes $\mathbb{E}_t u(a, D_m Y_{t+1})$.*

(i) (Step 2 is the operator.) *For every m, the first-order condition of step (2) is the behavioral condition $\mathbb{E}_t^{\mathrm{BR}} \nabla_a u(a, Y_{t+1}) = 0$ with $\rho_X = m$.*

(ii) (Quadratic loss.) *Let $Y_{t+1} = \varsigma \tilde{y}_{t+1}$, with $\mu_t \equiv \mathbb{E}_t \tilde{y}_{t+1}$ bounded and $\tilde{y}_{t+1}$ having conditionally bounded third moments. Let a_0 be the unique maximizer of $u(\cdot, 0)$, with $H \equiv \nabla^2_{aa} u(a_0, 0)$ negative definite, and let $U \equiv \nabla^2_{ay} u(a_0, 0)$. Read $a(m)$ as the local maximizer branch that converges to a_0 as $\varsigma \to 0$; it is the global maximizer under additional conditions, e.g. uniform strict concavity of u in a on a convex action domain. Then the loss from inattention is*

$$L_t(m) \equiv \mathbb{E}_t u\big(a(\mathbf{1}), Y_{t+1}\big) - \mathbb{E}_t u\big(a(m), Y_{t+1}\big) = \tfrac{1}{2}\varsigma^2 (\mathbf{1}-m)^\top \Lambda_t (\mathbf{1}-m) + o(\varsigma^2), \quad \text{(OA.10)}$$

with $\Lambda_t = \mathrm{diag}(\mu_t)\, W\, \mathrm{diag}(\mu_t)$ and $W \equiv U^\top (-H)^{-1} U \succeq 0$.

(iii) (Quadratic attention cost, attention chosen once.) *If attention is chosen once, before the state is known, to minimize $\mathbb{E}\left[\tfrac{1}{2}(\mathbf{1}-m)^\top \Lambda_t (\mathbf{1}-m)\right] + \sum_i \kappa_i m_i^2$ with $\kappa_i > 0$ (the*

leading-order loss, with ς^2 absorbed into κ), the minimizer over $[0,1]^n$ is unique. With $\bar{\Lambda} \equiv W \odot \mathbb{E}[\mu_t \mu_t^\top] \succeq 0$, if $\bar{\Lambda}$ is diagonal it is

$$\rho_{X_i} = m_i^* = \frac{\bar{\Lambda}_{ii}}{\bar{\Lambda}_{ii} + 2\kappa_i} \in [0,1). \tag{OA.11}$$

With a linear cost $\kappa_i m_i$ instead (Gabaix's sparsity-inducing choice), $m_i^* = \max\{0, 1 - \kappa_i/\bar{\Lambda}_{ii}\}$ for $\bar{\Lambda}_{ii} > 0$ and $m_i^* = 0$ if $\bar{\Lambda}_{ii} = 0$. (Uniqueness is for given stakes; the equilibrium-attention fixed point need not be unique.)

(iv) (Risk is not priced by step 1.) *The leading-order loss* (OA.10) *depends on the perception rule only through its conditional mean: any rule $Y^s = \varsigma \tilde{y}^s$ that shrinks with the perturbation scale, with uniformly bounded conditional third moments and $\mathbb{E}_t Y^s = D_m \mathbb{E}_t Y_{t+1}$, including the mean-only contraction $Y^s = D_m \mathbb{E}_t Y_{t+1} + (Y_{t+1} - \mathbb{E}_t Y_{t+1})$, has the same $L_t(m)$ up to $o(\varsigma^2)$. (Equal means alone do not suffice: a perception that does not shrink with ς can have the same mean and a loss of order one.)*

Proof of Proposition OA-A.4. (i) Step (2) maximizes $\mathbb{E}_t u(a, D_m Y_{t+1})$. Its first-order condition, $\mathbb{E}_t \nabla_a u(a, D_m Y_{t+1}) = 0$, is (OA.2) with $\mathcal{C}$ the chart contraction at $\rho_X = m$.

(ii) Let $F(a, \varsigma) \equiv \mathbb{E}_t \nabla_a u(a, \varsigma D_m \tilde{y}_{t+1})$. Bounded third derivatives of u and bounded conditional moments allow differentiation under the expectation, so F is C^1 near $(a_0, 0)$ with $F(a_0, 0) = 0$, $\partial_a F = H$ invertible and $\partial_\varsigma F = U D_m \mu_t$. The implicit function theorem gives a unique stationary point near a_0, a local maximizer since $H \prec 0$. This is the branch $a(m)$ of the proposition. (It is the global maximizer only under additional conditions that keep global maximizers near a_0 as $\varsigma \to 0$, such as uniform strict concavity on a convex action domain or a uniform tightness of the maximizers.) Along this branch, $a(m) = a_0 + \varsigma(-H)^{-1} U D_m \mu_t + o(\varsigma)$, hence $\Delta \equiv a(m) - a(\mathbf{1}) = \varsigma(-H)^{-1} U (D_m - I)\mu_t + o(\varsigma)$. Let $V(a) \equiv \mathbb{E}_t u(a, Y_{t+1})$, maximized at $a(\mathbf{1})$ with $\nabla V(a(\mathbf{1})) = 0$. A second-order Taylor expansion with bounded third derivatives gives

$$V(a(\mathbf{1})) - V(a(m)) = \tfrac{1}{2} \Delta^\top \big(- \nabla^2 V(a(\mathbf{1})) \big) \Delta + O(|\Delta|^3),$$

and $\nabla^2 V(a(\mathbf{1})) = H + O(\varsigma)$. Substituting Δ,

$$L_t(m) = \tfrac{1}{2} \varsigma^2 \mu_t^\top (I - D_m)\, U^\top (-H)^{-1} U\, (I - D_m) \mu_t + o(\varsigma^2),$$

and $(I - D_m)\mu_t = \mathrm{diag}(\mu_t)(\mathbf{1} - m)$ gives (OA.10). $W \succeq 0$ because $(-H)^{-1} \succ 0$.

(iii) $\mathbb{E}[(\mathbf{1} - m)^\top \Lambda_t (\mathbf{1} - m)] = \sum_{i,j} (1 - m_i)(1 - m_j) W_{ij} \mathbb{E}[\mu_{i,t} \mu_{j,t}] = (\mathbf{1} - m)^\top \bar{\Lambda} (\mathbf{1} - m)$, and $\bar{\Lambda} \succeq 0$ by the Schur product theorem. The objective is then strictly convex (the cost is strictly convex, the loss convex) and continuous on the compact convex set $[0,1]^n$, so

its minimizer is unique. If $\bar{\Lambda}$ is diagonal the problem separates; the first-order condition $-\bar{\Lambda}_{ii}(1-m_i)+2\kappa_i m_i=0$ gives (OA.11), which lies in $[0,1)$ and is therefore the constrained minimizer. With cost $\kappa_i m_i$, the separated objective $\frac{1}{2}\bar{\Lambda}_{ii}(1-m_i)^2+\kappa_i m_i$ has unconstrained minimizer $1-\kappa_i/\bar{\Lambda}_{ii}<1$, truncated at 0.

(iv) For a rule $Y^s=\varsigma\tilde{y}^s$ with uniformly bounded conditional third moments, the argument of (ii) applies with $\tilde{y}^s$ in place of $D_m\tilde{y}_{t+1}$, and the perception rule enters only through $\partial_\varsigma F$, that is, through $\mathbb{E}_t\nabla^2_{ay}u(a_0,0)\,\tilde{y}^s=U\,\mathbb{E}_t\tilde{y}^s$. Any two such rules with the same conditional mean therefore have the same $a(m)$ to first order in ς and the same $L_t(m)$ to second order. The small-noise restriction matters: on the action set $(-2,2)$, with $u(a,y)=-a^2/2+a(1-\cos y)$ and true $Y=0$, a perception equal to ±1 with equal probability has mean zero but chooses $a=1-\cos 1$, a loss of $(1-\cos 1)^2/2$ that does not vanish with ς. □

Scope of the sparse-max reading. Proposition OA-A.4 rationalizes the *form* of the operator (a chart-linear contraction toward a default placed at the reference point) and gives ρ an interpretation as optimal attention. It does so under three approximations, each of which should be read as a limitation. First, attention is set against Gabaix's leading-order quadratic loss. At higher order the stakes can also depend on Σ_t and higher cumulants, so the leading-order formula (OA.11) need not remain exact. Whether those higher moments enter the stakes depends on the decision problem: with $u(a,y)=-(a-y)^2/2$, the loss from $a=m\mathbb{E}_t y$ is exactly $(1-m)^2(\mathbb{E}_t y)^2/2$, and the variance cancels. Attention chosen once is still a constant parameter. Second, attention is chosen once; chosen state by state, it would depend on $\mu_t\mu_t^\top$ (Remark 7). Third, $\bar{\Lambda}$ depends on the equilibrium distribution of μ_t, which itself depends on ρ, so (OA.11) is a fixed-point condition, not a closed form. The reading also assumes that the agent treats Y_{t+1} parametrically. This covers pricing conditions in aggregate variables. For an Euler equation in the agent's own future consumption it needs the sparse dynamic-programming extension of Gabaix (2016a), which we do not reproduce here. With diagonal stakes, a quadratic cost ($\alpha=2$ in Gabaix's family) gives interior attention whenever $\bar{\Lambda}_{ii}>0$, and zero attention if $\bar{\Lambda}_{ii}=0$. With correlated stakes, it can give zero attention even when every diagonal stake is positive: with $\bar{\Lambda}=\left(\begin{smallmatrix}1 & -3\\ -3 & 9\end{smallmatrix}\right)$ and $\kappa_1=\kappa_2=3$, the minimizer is $m^*=(0,0.4)$, where the derivative in m_2 vanishes and the derivative in m_1 at $m_1=0$ is $0.8>0$. The linear cost gives exact myopia ($\rho=0$) for low-stakes variables, including those with $\bar{\Lambda}_{ii}=0$. With diagonal stakes, in both cases $\rho_i=1$ only in the limit $\kappa_i\to 0$, which is the natural reading of variables not selected for behavioral treatment. With correlated stakes full attention can occur at a positive cost: for $\bar{\Lambda}=\left(\begin{smallmatrix}1 & 2\\ 2 & 4\end{smallmatrix}\right)$ and $(\kappa_1,\kappa_2)=(0.1,1)$ the quadratic-cost minimizer is $m^*=(1,2/3)$ (the derivative in m_1 is $-7/15<0$ at the upper bound, the derivative in m_2 vanishes, and the Hessian is positive definite).

OA-B Risk and integrability

This section proves the results of Sections 2.4 and 2.5 of the main text and verifies the risk channel numerically. The statements are in the main text and are not repeated; each proof first lists the assumptions it uses.

OA-B.1 Proof of Proposition 2

Assumptions. The reference point $\bar{X}$ is fixed. $G(\hat{x}) \equiv g(\bar{X} \odot e^{\hat{x}})$ (with mixed charts, $G(\hat{x}) \equiv g(\chi^{-1}(\chi(\bar{X})+\hat{x}))$; nothing below changes) is three times continuously differentiable with third derivatives bounded in norm by $Ce^{c\|\hat{x}\|}$ for constants $C, c \geq 0$. (Bounded third derivatives are the case $c = 0$. A power function $g(X) = X^a$ gives $G(\hat{x}) = \bar{X}^a e^{a\hat{x}}$, whose third derivative is $a^3 \bar{X}^a e^{a\hat{x}}$, so $c = |a|$.) The forward deviation is $\hat{x}_{t+1} = \varsigma \tilde{x}_{t+1}$ with $\mathbb{E}_t\left[\|\tilde{x}_{t+1}\|^3 e^{\bar{c}\|\tilde{x}_{t+1}\|}\right] < \infty$ for some $\bar{c} > 0$ (bounded third moments suffice when $c = 0$).

Proof. By (6), $\mathbb{E}_t g(X_{t+1}) = \mathbb{E}_t G(\hat{x}_{t+1})$ and $\mathbb{E}_t^{\mathrm{BR}} g(X_{t+1}) = \mathbb{E}_t G(D_\rho \hat{x}_{t+1})$. A second-order Taylor expansion of G about $\hat{x} = 0$ gives $G(z) = G(0) + \nabla^\top z + \frac{1}{2} z^\top H z + R(z)$ with $\nabla = \nabla G(0)$, $H = \nabla^2 G(0)$. By Taylor's theorem with the integral form of the remainder, $|R(z)| \leq \frac{C}{6}\|z\|^3 e^{c\|z\|}$. Substitute $z = D_\rho \hat{x}_{t+1} = \varsigma\, D_\rho \tilde{x}_{t+1}$; since $\|D_\rho\| \leq 1$, for $\varsigma c \leq \bar{c}$,

$$\mathbb{E}_t|R(z)| \leq \tfrac{C}{6}\varsigma^3\, \mathbb{E}_t\left[\|\tilde{x}_{t+1}\|^3 e^{\bar{c}\|\tilde{x}_{t+1}\|}\right] = O(\varsigma^3) = o(\varsigma^2).$$

Taking $\mathbb{E}_t$ of the expansion,

$$\mathbb{E}_t^{\mathrm{BR}} g(X_{t+1}) = G(0) + \nabla^\top D_\rho\, \mathbb{E}_t \hat{x}_{t+1} + \tfrac{1}{2}\, \mathbb{E}_t\left[(D_\rho \hat{x}_{t+1})^\top H (D_\rho \hat{x}_{t+1})\right] + o(\varsigma^2).$$

Using $\mathbb{E}_t[z^\top H z] = (\mathbb{E}_t z)^\top H (\mathbb{E}_t z) + \mathrm{tr}(H \mathrm{Var}_t z)$ with $\mathbb{E}_t z = D_\rho \mu$ and $\mathrm{Var}_t z = D_\rho \Sigma D_\rho$, and $G(0) = g(\bar{X})$, yields (12). Equation (13) is $\mathrm{tr}(H D_\rho \Sigma D_\rho) = \sum_{i,j} H_{ij} (D_\rho \Sigma D_\rho)_{ji} = \sum_{i,j} \rho_{X_i} \rho_{X_j} H_{ij} \Sigma_{ij}$. Setting $\rho_X = \mathbf{1}$ recovers the rational second-order term $\frac{1}{2}\mathrm{tr}(H\Sigma)$, so the behavioral-to-rational ratio of the loading on Σ_{ij} is exactly $\rho_{X_i}\rho_{X_j}$; with $D_\rho = \rho I$ the whole term is multiplied by ρ^2. The heterogeneous-attention example in the main text is the case $G(z) = \frac{1}{2}(z_1 - z_2)^2$, $H = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, $\Sigma = \mathrm{Var}_t(u) \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, for which $\frac{1}{2}\mathrm{tr}(H\Sigma) = 0$ and, with $D_\rho = \mathrm{diag}(1, \frac{1}{2})$, $\frac{1}{2}\mathrm{tr}(H D_\rho \Sigma D_\rho) = \frac{1}{2}(1 - 1 + \frac{1}{4})\mathrm{Var}_t(u) = \frac{1}{8}\mathrm{Var}_t(u)$. □

OA-B.2 Proof of Corollary 1, and a non-Gaussian example

Assumptions. Fixed reference point $\bar{c}$; $\log c_{t+1}$ conditionally Gaussian with mean deviation $\mu_{c,t}$ and variance $\sigma^2_{c,t}$.

Proof. Apply $g(c) = c^{-1}$. Conditional Gaussianity of $\hat{c}_{t+1}$ makes the shrunk variable

$\rho\hat{c}_{t+1}$ Gaussian with mean $\rho\mu_{c,t}$ and variance $\rho^2\sigma^2_{c,t}$, so $\mathbb{E}^{\mathrm{BR}}_t c^{-1}_{t+1} = \bar{c}^{-1}\mathbb{E}_t e^{-\rho\hat{c}_{t+1}} = \bar{c}^{-1}\exp\big(-\rho\mu_{c,t}+\frac{1}{2}\rho^2\sigma^2_{c,t}\big)$ exactly (in the lognormal case the log moment generating function terminates at its second cumulant; the second-order Taylor expansion of the level expectation itself is not exact). Taking logs of $R_t = \beta^{-1}c_t^{-1}/\mathbb{E}^{\mathrm{BR}}_t c^{-1}_{t+1}$, with $\hat{c}_t = \log c_t - \log\bar{c}$, gives (14). □

A non-Gaussian example. In an Aruoba-style law of motion, $\log c_{t+1} = c_0 + \psi\log c_t + \psi_2 s_t^2 + (1+\gamma s_t)\sigma_1 u_{t+1} + \sigma_2 u^2_{t+1}$ with $u_{t+1}\sim N(0,1)$, conditional Gaussianity fails (u^2 is chi-squared), so (14) holds to second order (Proposition 2) rather than exactly. From $\mathbb{E}_t u_{t+1} = 0$, $\mathbb{E}_t u^2_{t+1} = 1$, $\mathrm{Var}_t(u^2_{t+1}) = 2$ and $\mathbb{E}_t u^3_{t+1} = 0$, the conditional variance is $\sigma^2_{c,t} = (1+\gamma s_t)^2\sigma_1^2 + 2\sigma_2^2$, and the state-independent *component* of the precautionary wedge is $-\frac{1}{2}\rho^2(\sigma_1^2+2\sigma_2^2)$.

OA-B.3 Proof of Proposition 3

Assumptions. $g(X) = X^a$; X a scalar log-chart variable with attention ρ and fixed reference point $\bar{X}$; $\mathcal{M}_t(s) = \mathbb{E}_t e^{s\hat{x}_{t+1}} \in (0,\infty]$. For the series part, $\mathcal{M}_t$ is finite in a neighborhood of zero, so the cumulant series $\psi_t(s) = \sum_{k\geq 1}\kappa_{k,t}s^k/k!$ has a positive radius of convergence r_t, and $|a|\rho < r_t$.

Proof. By Definition 1, the perceived level is $\bar{X}e^{\rho\hat{x}_{t+1}}$, so $\mathbb{E}^{\mathrm{BR}}_t X^a_{t+1} = \mathbb{E}_t\big(\bar{X}e^{\rho\hat{x}_{t+1}}\big)^a = \bar{X}^a\,\mathbb{E}_t e^{a\rho\hat{x}_{t+1}} = \bar{X}^a\mathcal{M}_t(a\rho)$. The integrand is positive, so the identity holds in $[0,\infty]$ whether or not either side is finite; this is (15). On $|s| < r_t$ the series converges and equals $\log\mathcal{M}_t(s)$, so for $|a|\rho < r_t$, $\mathcal{M}_t(a\rho) = \exp\big(\sum_k \kappa_{k,t}(a\rho)^k/k!\big)$, which is (16); the k-th term carries $(a\rho)^k = \rho^k a^k$, i.e. ρ^k times the rational term $\kappa_{k,t}a^k/k!$. The perceived deviation $\rho\hat{x}_{t+1}$ has cumulants $\rho^k\kappa_{k,t}$, so for $\rho > 0$ $\rho^3\kappa_3/(\rho^2\kappa_2)^{3/2} = \kappa_3/\kappa_2^{3/2}$ and $\rho^4\kappa_4/(\rho^2\kappa_2)^2 = \kappa_4/\kappa_2^2$. In the level chart the perceived deviation is $\rho\hat{x}_{t+1}$ for any law, so each of its cumulants that exists is $\rho^k\kappa_{k,t}$, with no series involved. □

OA-B.4 Proof of Corollary 2, and the threshold itself

Assumptions. As in Proposition 3, with $\mathbb{E}_t X^a_{t+1} = \infty$ and $\alpha_t^+ > 0$ if $a > 0$, $\alpha_t^- < 0$ if $a < 0$.

Proof. The set $\{s : \mathcal{M}_t(s) < \infty\}$ is an interval containing 0 (by convexity of $s\mapsto e^{sx}$), with endpoints $\alpha_t^- \leq 0 \leq \alpha_t^+$. Divergence of the rational expectation means $\mathcal{M}_t(a) = \infty$, which implies $a \geq \alpha_t^+$ if $a > 0$, or $a \leq \alpha_t^-$ if $a < 0$. For $a > 0$, $\rho < \alpha_t^+/a$ gives $0 \leq a\rho < \alpha_t^+$, and for $a < 0$, $\rho < \alpha_t^-/a$ gives $\alpha_t^- < a\rho \leq 0$. For $\rho > 0$, $a\rho$ lies strictly between zero and the relevant abscissa, inside the interval of finiteness; at $\rho = 0$, $\mathcal{M}_t(0) = 1$, which is finite even when zero is an endpoint of that interval. Either way, by (15) $\mathbb{E}^{\mathrm{BR}}_t X^a_{t+1} = \bar{X}^a\mathcal{M}_t(a\rho)$ is finite. The threshold $\bar{\rho}_t$ is strictly positive because the relevant abscissa is nonzero. The series statement is Proposition 3; the bound $r_t \leq \min\{\alpha_t^+, -\alpha_t^-\}$ holds because the series converges on $|s| < r_t$ to a function analytic there, while the real abscissa of convergence

of the Laplace transform of a positive measure is a singular point of the transform. If the relevant abscissa is zero, as for a Student-t deviation, whose moment generating function is infinite at every $s \neq 0$, then $\mathcal{M}_t(a\rho) = \infty$ for every $\rho > 0$. □

At the threshold. At $\rho = \bar{\rho}_t$ exactly, $a\rho$ equals the abscissa, and finiteness depends on whether the moment generating function is finite there. In the exponential example of Section 4.2, with $\log X \sim \text{Exp}(\lambda)$ and $\bar{X} = 1$, $\mathcal{M}_t(s) = \lambda/(\lambda - s)$ for $s < \lambda$, so $\bar{\rho}_t = \lambda/a$. The behavioral expectation $\lambda/(\lambda - a\rho)$ then diverges as $\rho \uparrow \bar{\rho}_t$. By contrast, for a law with density proportional to $e^{-\lambda x}/(1 + x^2)$ on $x \geq 0$, the moment generating function is finite at $s = \lambda$, and so is the behavioral expectation at the threshold. Corollary 2 therefore uses a strict inequality.

OA-B.5 Verifying the risk-discounting channel

Proposition 2 and its Corollary 1 predict that cognitive discounting scales the loading on conditional *second* moments by the square of attention: with a common ρ, the order-2 precautionary/risk correction is attenuated by ρ^2 relative to rational expectations, and is identically zero under certainty equivalence. We verify this directly in the Aruoba-style Euler equation with Gaussian log-consumption,

$$\log R_t = -\log \beta - \widehat{c}_t + \rho\, \mu_{c,t} - \tfrac{1}{2}\, \rho^2\, \sigma^2_{c,t},$$

whose precautionary term $-\frac{1}{2}\rho^2\sigma_c^2$ is the object of the Corollary. Figure OA.1 and Table OA.1 plot this precautionary term against ρ. The closed form and a four-million-draw Monte-Carlo evaluation of the same behavioral expectation agree to $O(10^{-4})$ in the reported table. The closed form has log–log slope exactly 2; the Monte-Carlo points, as reported, give 1.99. (The Monte Carlo checks the closed-form expression, not the solved model.) The wedge scales as ρ^2. Under certainty equivalence (a first-order solution) the wedge is identically zero for every ρ.

Two remarks connect this to the quantitative model. First, the risk channel is precisely the novelty that certainty-equivalent (first-order) methods cannot express, which is why the model comparison of §5 — conducted at first order — cannot see it. Second, in the full laboratory model the same contraction of forward-looking terms is the candidate mechanism (Remark 5) behind the simulation evidence that, for low enough attention, it removes explosive higher-order paths when divergence is driven by persistence (§4.2).

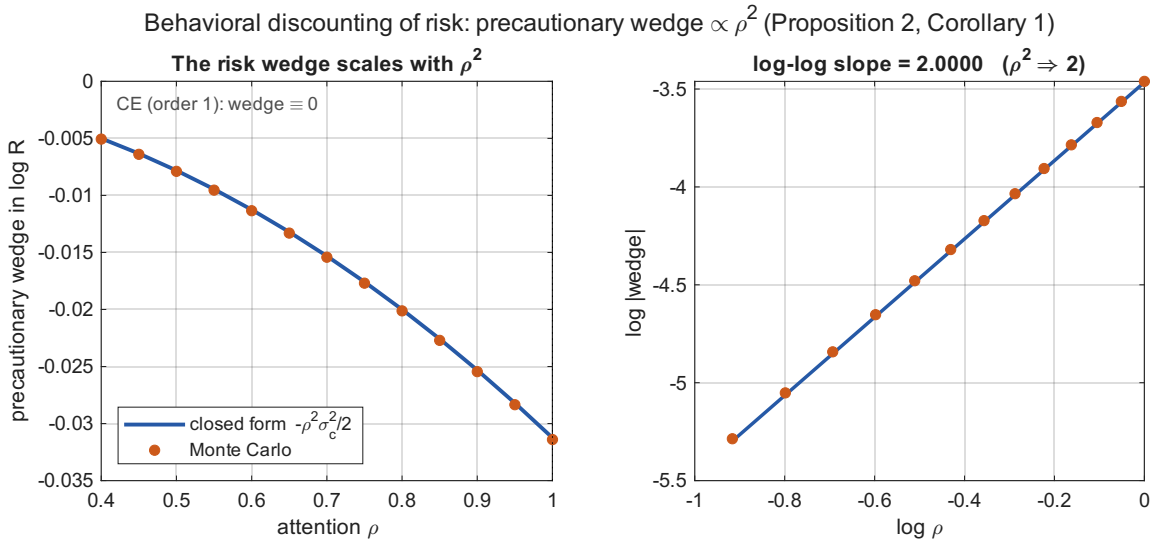


Figure OA.1: Behavioral discounting of risk (Proposition 2, Corollary 1). The precautionary wedge $-\frac{1}{2}\rho^2\sigma_c^2$ in the real rate against attention ρ: closed form versus a Monte-Carlo behavioral expectation (left), and on a log–log scale (right), where the closed-form slope is 2 (Monte Carlo: 1.99) — the wedge scales as ρ^2. Under certainty equivalence the wedge is identically zero.

Table OA.1: Precautionary wedge $-\frac{1}{2}\rho^2\sigma_c^2$ by attention ρ: closed form versus Monte Carlo.

precautionary wedge $-\frac{1}{2}\rho^2\sigma_c^2$	$\rho = 1.0$	$\rho = 0.9$	$\rho = 0.8$	$\rho = 0.7$	$\rho = 0.6$	$\rho = 0.5$
closed form	-0.0312	-0.0253	-0.0200	-0.0153	-0.0113	-0.0078
Monte Carlo	-0.0314	-0.0254	-0.0201	-0.0154	-0.0113	-0.0079

OA-C Trends and regimes

OA-C.1 The moving reference point

Let X_i be a log-chart level variable that grows, and write $X_{i,t+1} = T_{i,t+1}\, x_{i,t+1}$, where $T_{i,t+1}$ is its trend and $x_{i,t+1}$ is stationary. The trend is an anchor state in the sense of §OA-A.2: it is known at $t+1$ (it may be known already at t) and is perceived correctly. Set the reference point to the trend times the steady state of the detrended variable, $\bar{X}_{i,t+1} = T_{i,t+1}\bar{x}_i$. Then (6) gives

$$\log \tilde{X}_{i,t+1} = \log \bar{X}_{i,t+1} + \rho_i\big(\log X_{i,t+1} - \log \bar{X}_{i,t+1}\big) = \log T_{i,t+1} + \log \bar{x}_i + \rho_i\big(\log x_{i,t+1} - \log \bar{x}_i\big),$$

that is, $\tilde{X}_{i,t+1} = T_{i,t+1}\,\tilde{x}_{i,t+1}$ with $\tilde{x}_{i,t+1} = \bar{x}_i(x_{i,t+1}/\bar{x}_i)^{\rho_i}$ the contraction of the detrended variable toward its steady state. Every contracted level is the trend times the contracted detrended variable, and the trend enters exactly as it does in the rational model. Hence, if the rational equilibrium conditions can be written in detrended variables by dividing by the trends (as on a balanced-growth path), so can the behavioral ones, with each forward detrended variable contracted toward its steady state. The levels model and its detrended twin then have the same equilibrium conditions, and the same likelihood when the observables are the same. In the laboratory model (Appendix A), the trend of the real quantities is realized technology, $T_{t+1} = A_{t+1}$. The trend of the price level is the lagged price level grown at trend inflation, $T_{t+1} = P_t(1+\pi^*)$, so the contracted price level is the contraction of gross inflation relative to its target. Numerically, the levels model was checked against a detrended twin under rational expectations (§OA-G.9); the twin in the code detrends the price level differently, so it is not used under cognitive discounting.

Because the trend is the anchor, it is not contracted: the risk in T_{t+1} is perceived correctly, and the cumulant results of Section 2.5 apply to the detrended deviations conditional on the trend (qualification (a) in §2.5).

OA-C.2 Proof of Proposition 4

Assumptions. A Markov chain $r_t \in \{1,\dots,h\}$ with transition probabilities $\pi_{r_t,j}(X_t)$ measurable in X_t and summing to one over j; regime-specific attention $\rho_X(r_t)$, reference points $\bar{X}(j)$ and parameters $\theta(r_t)$.

Proof. Equation (17) applies the chart contraction (4) inside each continuation term of the rational switching system $0 = \mathbb{E}_t \sum_j \pi_{r_t,j} f(X_{t+1}(j),\dots,\theta(r_t))$. Since the contraction is a continuous, hence measurable, map for each (r_t, j), the transformed system inherits the

measurability of its equilibrium conditions from the rational one and is a switching system of the same form. Existence and determinacy of a solution are not implied and must be checked case by case. Nesting (ii): at $\rho_X = \mathbf{1}$ the contraction is the identity map term by term, so (17) *is* the rational switching system — an identity on equations, hence exact at every order of approximation. Nesting (i): with ρ_X, $\bar{X}$, θ regime-invariant, every summand carries the same transformation and the same parameters, and (17) collapses to (5) — again an identity on equations. □

OA-D Analytical applications

This section collects the algebra behind the three-equation comparisons with Gabaix (2020) in Sections 4.1 and 4.4 of the main text. The model is (10)–(11). To nest Gabaix's IS curve, write the coefficient on expected inflation in the real rate as b_π: $b_\pi = m$ for the operator, $b_\pi = 1$ for Gabaix. In this section, as in Table OA.2, $M \equiv m$ denotes Gabaix's attention, as in his paper. His M is a coefficient on expected future variables, that is, the term-wise rule of §2.1 applied to a linear model (see §5.7.4). His firm coefficient is set to $M^f = M$ (Remark 2). Throughout, $\beta \in (0,1)$, $\kappa, \sigma > 0$, $m \in [0,1]$ and $\phi_\pi, \phi_x \geq 0$.

OA-D.1 Determinacy and the permanent peg

With the Taylor rule $i_t = \phi_\pi \pi_t + \phi_x x_t$ (the natural rate plays no role in determinacy), stack $z_t = (x_t, \pi_t)'$ and write the system as $\mathcal{A} z_t = \mathcal{B}\, \mathbb{E}_t z_{t+1}$, with

$$\mathcal{A} = \begin{pmatrix} 1 + \sigma\phi_x & \sigma\phi_\pi \\ -\kappa & 1 \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} m & \sigma b_\pi \\ 0 & \beta m \end{pmatrix}.$$

Both variables are jumps, so the equilibrium is determinate if and only if both eigenvalues of the forward matrix $F \equiv \mathcal{A}^{-1}\mathcal{B}$ lie inside the unit circle. With $\Delta \equiv \det \mathcal{A} = 1 + \sigma\phi_x + \kappa\sigma\phi_\pi > 0$,

$$F = \frac{1}{\Delta} \begin{pmatrix} m & \sigma b_\pi - \sigma\phi_\pi \beta m \\ \kappa m & \kappa\sigma b_\pi + (1 + \sigma\phi_x)\beta m \end{pmatrix},$$

so that

$$\det F = \frac{\beta m^2}{\Delta}, \qquad \mathrm{tr} F = \frac{m + \kappa\sigma b_\pi + (1 + \sigma\phi_x)\beta m}{\Delta}.$$

For a real 2×2 matrix both eigenvalues lie inside the unit circle if and only if $|\det F| < 1$ and $|\mathrm{tr} F| < 1 + \det F$. With nonnegative ϕ_π, ϕ_x the first condition and the lower half of the second hold, and the binding condition $1 + \det F > \mathrm{tr} F$ reads $\Delta + \beta m^2 > m + \kappa\sigma b_\pi + (1 + \sigma\phi_x)\beta m$. Using $1 - m - \beta m + \beta m^2 = (1 - m)(1 - \beta m)$ and dividing by $\kappa\sigma$,

$$\phi_\pi + \frac{1 - \beta m}{\kappa}\, \phi_x + \frac{(1 - \beta m)(1 - m)}{\kappa\sigma} > b_\pi. \tag{OA.12}$$

With $b_\pi = 1$ this is Gabaix's Proposition 3.1 with $M^f = M$, $\phi_\pi + \frac{1-\beta M}{\kappa}\phi_x + \frac{(1-\beta M)(1-M)}{\kappa\sigma} > 1$; with $b_\pi = m$ it is the operator's criterion, which has m rather than 1 on the right-hand side and is therefore easier to satisfy.

The permanent peg. Under a peg ($\phi_\pi = \phi_x = 0$) only the last term of (OA.12) remains. At Gabaix's calibration ($\beta = 0.99$, $\kappa = 0.11$, $\sigma = 0.2$) it equals 1.08 at $M = 0.85$, so his economy is determinate even under a permanent interest-rate peg; the peg is determinate for $M \lesssim 0.856$. With his originally reported compounded firm coefficient, $M^f \approx 0.80$ at $M = 0.85$ (his formula gives 0.786 at a Calvo parameter of 0.75; Table OA.2), the term is larger, 1.42, and letting M^f vary with M as in his formula the cutoff is $M \lesssim 0.88$. Our calibration ($\beta = 0.99$, $\kappa = 0.10$, $\sigma = 1$) gives a determinate peg only for $m \lesssim 0.733$ under the operator, and for $m \lesssim 0.687$ with Gabaix's IS curve. The difference between the two calibrations comes mainly from the intertemporal elasticity σ; the difference between the two IS curves is the operator's discounting of expected inflation in the real rate. (The cutoffs with $M^f = M$ and for the operator are computed from the eigenvalues of the pegged forward matrix of §OA-D.2, the compounded-coefficient cutoff from the peg term of his Proposition 3.1; with $M^f = M$ the two methods give the same cutoff, as (OA.12) implies.) Table OA.2 collects these values.

Table OA.2: Three-equation determinacy and forward-guidance comparisons (analytic). Our calibration: $\beta = 0.99$, $\kappa = 0.10$, $\sigma = 1$; Gabaix's: $\beta = 0.99$, $\kappa = 0.11$, $\sigma = 0.2$. Peg: nominal rate fixed. The compounded cutoff varies between 0.876 and 0.881 for Calvo parameters between 0.5 and 0.9.

largest attention with a determinate peg, our calibration, operator	0.7329
same, Gabaix IS curve (inflation in the real rate undiscounted)	0.6872
same, Gabaix calibration, Gabaix IS curve	0.8559
forward eigenvalue, Gabaix calibration, $M = 0.85$	0.9937
forward eigenvalue, Gabaix calibration, $M = 0.90$	1.0474
Gabaix Prop. 3.1 peg term at $M = 0.85$, $M^f = M$	1.0807
compounded firm coefficient M^f at $M = 0.85$ (Calvo $\theta = 0.75$)	0.7858
Gabaix Prop. 3.1 peg term at $M = 0.85$, $M^f = 0.80$	1.4182
largest M with peg term > 1, $M^f = M$	0.8559
largest M with peg term > 1, compounded M^f	0.8805

OA-D.2 The forward matrix and the forward-guidance puzzle

With the nominal rate exogenous, $\mathcal{A}$ is lower triangular with unit diagonal and the system is $z_t = F\,\mathbb{E}_t z_{t+1} + G\,i_t$ with

$$F = \begin{pmatrix} m & \sigma b_\pi \\ \kappa m & \beta m + \kappa\sigma b_\pi \end{pmatrix}, \qquad G = -\sigma \begin{pmatrix} 1 \\ \kappa \end{pmatrix}.$$

A one-period cut announced T quarters ahead moves z today by $F^T G$ times the cut, so the long-horizon effect is governed by the dominant eigenvalue of F. For the operator ($b_\pi = m$), $F = m\,F_1$ with $F_1 = \left(\begin{smallmatrix} 1 & \sigma \\ \kappa & \beta+\kappa\sigma \end{smallmatrix}\right)$ the rational forward matrix: attention scales every eigenvalue by m. At our calibration the dominant eigenvalue of F_1 is 1.364, so announced cuts decay with the horizon once m is below its reciprocal, 0.733. This is the peg-determinacy cutoff of §OA-D.1: with a pegged rate only the forward matrix remains, so the two thresholds are the same object. With Gabaix's IS curve ($b_\pi = 1$) the matrix is not proportional to m; at his calibration its dominant eigenvalue is 0.99 at $M = 0.85$ and 1.05 at $M = 0.90$, below one only for $M \lesssim 0.856$. Gabaix (2020) (his Figure 2, after McKay et al., 2016) instead cuts the *real* rate. Then $x_t = M\,\mathbb{E}_t x_{t+1} - \sigma r_t$ alone determines output, and the response today to a real-rate cut T quarters ahead is σM^T, which dies out with the horizon for any $M < 1$, whatever the Phillips-curve slope. With the nominal rate held fixed, as in the main text, inflation feeds back into the real rate and the threshold depends on κ and σ.

OA-E Estimation details

The likelihood and measurement error. The model is estimated in levels, solved at first order, with the diffuse component of the state (the unit roots) initialized by an exact diffuse prior and the lower bound handled by an approximate constraint-aware filter. With the measurement errors *fixed* at a numerical floor, a handful of crisis quarters (2008–2010) can receive disproportionate weight in the likelihood. Small measurement error does not by itself make the likelihood degenerate — the structural shocks still leave forecast-error variance — and the sensitivity may reflect small forecast-error variances, misspecification, or numerical problems. All reported results estimate the measurement errors.

Optimization. The likelihood of the behavioral models has several local peaks, so a derivative-based optimizer started from the prior mean is not reliable for them. It can leave them with a *lower* maximized log-likelihood than the rational model they nest, which can only be an optimization failure. The nesting inequality concerns the maximized likelihood, or its supremum when the rational model sits on the boundary $\rho = 1$. It does not apply to the posterior density at the mode or to the marginal likelihood, whose priors and dimensions differ. We therefore use a global optimizer (an artificial bee colony), warm-started along the nesting chain (rational expectations, then a single attention, then the sector-, block- and variable-specific versions), and polish each mode with a derivative-based optimizer from several dispersed starts.

Posterior sampling. Posterior draws are generated by the Metropolis–Hastings algorithm in logit-transformed parameters (all parameter bounds are finite), which lets parameters near a bound mix properly. For the rational model and the common-attention restriction under the Beta prior, the proposal is a random walk whose covariance is the inverse finite-difference Hessian at the mode, re-tuned on a pilot run. The flat-prior attention specifications of Table 11 have posteriors with several local modes in non-attention parameters. For them, the proposal is a fixed mixture of three moves: a local random walk, an independence proposal drawn from a Student-t mixture fitted to preliminary posterior draws, and mode-jump moves. Four chains are started from dispersed points and the first iterations are discarded as warm-up. Convergence is assessed with the rank-normalized split-$\hat{R}$ and the bulk and tail effective sample sizes of Vehtari et al. (2021); the standard is $\hat{R} < 1.01$ and both effective sample sizes ≥ 400.

OA-E.1 Variable-by-variable attention: details

What is estimated. Each discounted variable $X \in \{C, \Lambda, P, Y, \pi, \pi^w, \tilde{\pi}, \tilde{\pi}^w\}$ has its own attention $\rho_X \in [0,1]$, on the same nonstationary model, data and lower-bound filter for every specification of Table 11. Every attention parameter gets a Uniform$(0,1)$ prior, and the restrictions are estimated under the same flat prior, so that the marginal likelihoods in that table share a prior family; all other priors are those of Appendix A. Each specification is sampled with four chains. Table OA.3 gives the full results behind Table 11: the number of sampled and fixed attention parameters, the log posterior and log-likelihood at the posterior mode, the sampled marginal data densities with their batch-dispersion diagnostics, and the convergence diagnostics. Its last row gives the posterior mode of the variable-specific model under a Beta$(0.85, 0.10)$ prior on every attention, at which all eight attentions are interior (between 0.85 and 0.97).

Table OA.3: Attention estimated variable by variable, and its restrictions, with flat priors: full results. The table reports the attention parameters sampled and fixed (for the last row, optimized only), the log posterior and log-likelihood at the posterior mode, and the sampled log marginal data density by bridge sampling and the modified harmonic mean. The mode lies on a bound of the unit interval for the richer models. The figures in parentheses (labeled NSE) are batch-dispersion diagnostics: the standard deviation of eight half-chain estimates (four chains times two halves) divided by $\sqrt{8}$, not a calibrated error of the pooled estimate (see §5.5). The table also gives the largest rank-normalized split-$\hat{R}$ and the smallest bulk and tail effective sample sizes over all parameters. "Converged": $\hat{R} < 1.01$ and both effective sample sizes ≥ 400. The first row is the baseline; the second imposes rational expectations on the two reference inflation rates; the common ρ is the robustness restriction of §5.7. The last row uses a Beta$(0.85, 0.10)$ prior and is not comparable with the others. The rows "Variable-specific, $\rho_{\tilde{\pi}} = \rho_{\tilde{\pi}^w} = 1$" and "Block-specific" have the same log posterior and log-likelihood at the mode: both optimizations reach the same point on bounds of the unit interval, where the two specifications give the same likelihood (the values agree to 10^{-9}). Away from it they differ, and so do their marginal data densities.

Specification	attention sampled/fixed	at the mode log post.	at the mode log L	sampled log MDD (NSE) bridge	sampled log MDD (NSE) MHM	max $\hat{R}$	min ESS bulk/tail	converged
Uniform$(0,1)$ *prior on every sampled attention parameter*								
Variable-specific	6/2	4308.46	4246.93	4189.93 (0.10)	4190.80 (0.13)	1.006	751/442	yes
Variable-specific, $\rho_{\tilde{\pi}} = \rho_{\tilde{\pi}^w} = 1$	4/4	4307.43	4245.42	4188.53 (0.12)	4188.87 (0.30)	1.009	878/668	yes
Households vs. firms	2/0	4305.08	4244.71	4190.29 (0.18)	4190.75 (0.21)	1.009	881/950	yes
Block-specific	4/0	4307.43	4245.42	4188.88 (0.18)	4189.17 (0.45)	1.009	831/461	yes
Common ρ	1/0	4302.49	4243.98	4188.50 (0.09)	4188.54 (0.10)	1.007	894/1242	yes
Beta$(0.85, 0.10)$ *prior on every attention parameter (mode only)*								
Variable-specific, Beta prior	– (eight optimized; no sampling)	4318.10	4245.17	–	–	–	–	–

A first-order identification property. The attentions on the price level and on output, ρ_P and ρ_Y, leave the likelihood exactly unchanged at first order. This is a property of the operator in this model. The operator contracts the deviation of a variable dated $t+1$ from

its reference point, so it moves the first-order solution only through the derivative of the forward-looking conditions with respect to that variable. In this model P_{t+1} and Y_{t+1} appear only in the forward terms of the wage and price Phillips curves, (A.7) and (A.13), where they multiply an indexation gap ($\Pi^w_{t+1}/\tilde{\Pi}^w_{t+1} - 1$ or $\Pi_{t+1}/\tilde{\Pi}_{t+1} - 1$) that is zero in the steady state. Those derivatives are therefore zero, and so is the first-order effect of ρ_P and ρ_Y; numerically, moving either over the unit interval leaves the log posterior unchanged. More generally, when the full forward derivative with respect to a variable vanishes in the steady state, attention to that variable is identified, if at all, only at second or higher order. This happens when the variable enters only multiplied by vanishing factors that do not themselves depend on it. The last condition matters: with reference point $X = 1$, $X(X-1)$ has a vanishing factor but derivative one. The priors on ρ_P and ρ_Y are proper, normalized and independent of the remaining parameters θ, and the two enter the likelihood through no other channel. They therefore integrate out with no evidence penalty: with $\eta = (\rho_P, \rho_Y)$, $L(y \mid \theta, \eta) = L(y \mid \theta)$ and $\pi(\theta, \eta) = \pi(\theta)\pi(\eta)$ give $\pi(\eta \mid y) = \pi(\eta)$ and $Z = \int L(y \mid \theta)\pi(\theta)\, d\theta$. They are therefore held fixed in sampling, which leaves the marginal likelihood unchanged; the other six are sampled, in logit-transformed space as above, because several of them have their mode on a bound.

Results. The Laplace approximation is not available for the variable-specific and block-specific models: their modes lie on bounds, where the Hessian is not positive definite. Posterior mean attentions are 0.89 for consumption, 0.76 for marginal utility, 0.91 for price inflation, 0.79 for wage inflation, and 0.52 and 0.45 for the two indexation rates (standard deviations and 95% HPD intervals in Table 11). The posterior means of the first four lie between 0.76 and 0.91; the two reference-inflation attentions have wide posteriors centered near one half, and fixing both at one lowers the log marginal data density by 1.40 (bridge).

Diagnostics. Every specification of Table 11 meets the convergence standard on all parameters: the largest $\hat{R}$ is 1.006 for the variable-specific model, 1.009 for the sector-specific and 1.009 for the block-specific version, and the smallest bulk or tail effective sample size is 442 across the five behavioral specifications and 428 with the rational-expectations row. The marginal likelihoods come from these draws; the spread between the two estimators (up to 0.9 log points, variable-specific) is a measure of sensitivity, not an error bound, and so is the batch dispersion in Table OA.3. In each of the five behavioral specifications, the pooled bridge estimate exceeds the mean of its half-chain estimates by 0.47 to 0.76 log points (§5.5).

OA-E.2 The common-attention restriction: details

This subsection collects the details of the robustness restriction of §5.7, a single attention ρ for all eight variables, which makes the variable-, sector- and block-specific operators coincide.

Under the flat prior. Under the Uniform$(0, 1)$ prior of the baseline estimation, the restriction has largest $\hat{R}$ 1.007 and smallest effective sample size 894 over all parameters (1145 for ρ; Table 11). The variable-specific model exceeds it by 1.43 log points (bridge; 2.3 by the modified harmonic mean), the sector-specific version by 1.8, and the block-specific version lies within 0.4 of it. At the mode, the variable-specific model's log posterior is 6.0 higher and its log-likelihood 2.95 higher; a difference in log posterior at the mode is not a model-selection statistic. The model has seven nominal extra attention parameters, of which five are likelihood-relevant directions once ρ_P and ρ_Y are integrated out. Under the Beta prior of §5.7, the bridge estimate of the restriction is 4190.6; the prior differs, so that number is not comparable with the rows of Table 11.

Posterior sampling. The rational model uses three chains of 220,000 kept draws, after a pilot run of 6,500 iterations; the restricted behavioral model four chains of 165,994, started at the global mode, at the second local mode, and at two re-optimized points of the ρ profile ($\rho = 0.80$ and 0.95). The posterior has two local modes: the global mode has $\rho = 0.897$, the other ($\rho = 0.917$) has government-spending persistence near its upper bound. All parameters of both posteriors meet $\hat{R} < 1.01$ and effective sample sizes ≥ 400: the largest $\hat{R}$ is 1.004 (RE) and 1.006 (CD), the smallest bulk effective sample size 1,130, and the smallest tail effective sample size 401 (labor-supply persistence, CD; 428 for wage-markup persistence, RE). For ρ, $\hat{R}$ is 1.002 and the bulk and tail effective sample sizes are 2,957 and 7,275. Both posteriors also have a third, rarely visited region, in which the wage markup is nearly a unit root and the persistence of the labor-supply shock is spread over a wide range. Measured as the draws with wage-markup persistence of at least 0.98, this region holds 1.1% of the RE draws and 1.3% of the CD draws. It is included in every statistic; excluding it leaves the posterior mean of ρ at 0.879. Draws with a non-finite likelihood (4,006) are excluded from every statistic. Of two further estimators not in Table 13, reciprocal importance sampling agrees (15.3); the Sims–Waggoner–Zha estimator agrees on the ranking but not on the magnitude (17.3, batch-dispersion diagnostic 1.3).

The posterior of ρ. Figure OA.2 and Table OA.4 show the posterior. It is interior and well determined, concentrated away from one; the largest of the 663,976 (autocorrelated) draws is 0.966, a descriptive statistic. The second, minor mode (about 4% of the mass)

differs from the main one in the persistence of government spending, of the preference shock, and of the labor-supply shock, not in attention: its mean ρ is 0.874 (its mode is at $\rho = 0.917$).

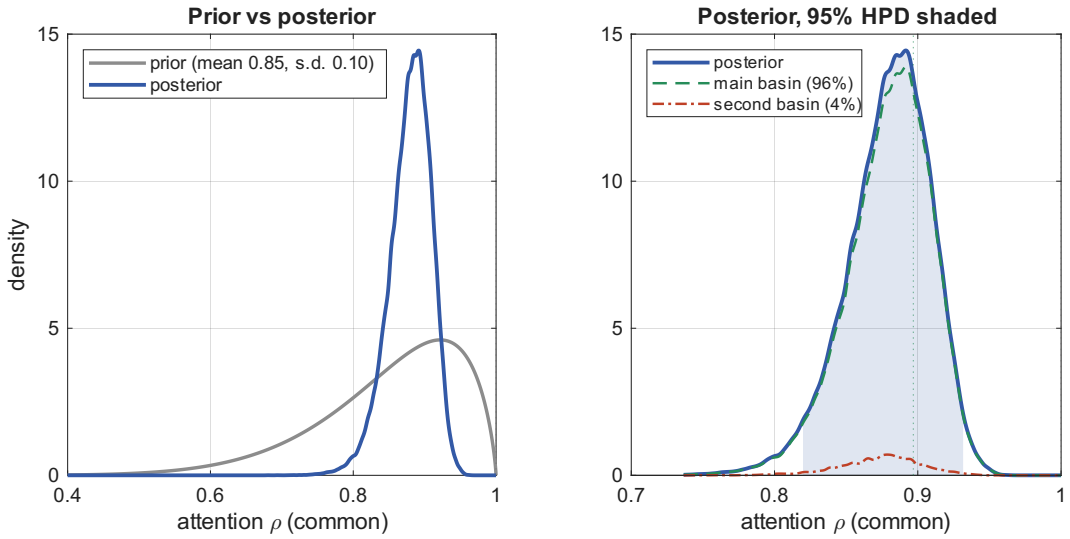


Figure OA.2: Common-attention restriction: posterior of ρ against its Beta$(0.85, 0.10)$ prior, pooled over four random-walk Metropolis chains.

Table OA.4: Common-attention restriction: posterior summary of ρ. Basins: draws assigned to the posterior mode or to the second local mode.

Posterior mean (sd)	0.879 (0.029)
Posterior median	0.882
95% credible set (equal-tailed)	[0.814, 0.928]
95% HPD interval	[0.820, 0.932]
$P(\rho \geq 0.95 \mid Y)$	0.001
$P(\rho \geq 0.99 \mid Y)$	0.000
Posterior mass, main basin (mean ρ)	0.96 (0.880)
Posterior mass, second basin (mean ρ)	0.04 (0.874)
Largest draw (663,976 draws)	0.966
Rank-normalized split $\hat{R}$ (4 chains)	1.002
Effective sample size, bulk	2957
Effective sample size, tail	7275

Prior sensitivity. Table OA.5 and Figure OA.3 report the re-estimation under four alternative priors (§5.7.2). What the prior changes is the marginal data density. A partial, ρ-specific prior and curvature term accounts for much of the change. It is the log prior density of ρ at the mode plus $\frac{1}{2}\log(2\pi s^2)$, with s the conditional posterior standard deviation of ρ. A diffuse prior spreads mass where the likelihood is negligible: under the uniform

prior the Laplace marginal data density falls by 1.4 log points, and this term alone accounts for 1.5. A tight prior placed away from the likelihood peak is penalized heavily: under Beta$(0.99, 0.01)$ the fall is 5.7, and this term alone accounts for 4.0. The remainder comes from the other parameters' priors and the rest of the Laplace covariance determinant, which this term does not capture. For the alternative priors the table reports posterior modes and Laplace approximations; sampled results in this table are those under the Beta$(0.85, 0.10)$ reference prior (the common restriction under the flat prior is sampled separately, Table 11). Under Beta$(0.99, 0.01)$, the best point found has a log-likelihood of 4248.8 at $\rho = 0.947$, with government-spending persistence on its upper bound. The Laplace approximation is invalid on a bound, so the table reports an interior evaluation point: the best interior point found, which need not be a stationary point.

Table OA.5: Common-attention restriction: sensitivity to the prior on ρ; all other priors unchanged. In this table, posterior draws and the bridge estimate for the Beta$(0.85, 0.10)$ prior only, from the chains of Table 13 (the flat-prior common restriction is sampled in Table 11); for the other priors, posterior mode and Laplace approximation (–: not reported). "reference": the Beta$(0.85, 0.10)$ prior of §5.7. Occam term: the partial, ρ-specific prior and curvature contribution, log prior density of ρ at the mode plus $\frac{1}{2}\log(2\pi s^2)$, s the conditional posterior standard deviation of ρ; it is not the full Laplace penalty, which also involves the other parameters' priors and the full covariance determinant. ‡Under Beta$(0.99, 0.01)$ the best point found has ρ_g on its upper bound, where the Laplace approximation is invalid; mode and Laplace are reported at the best interior point found (an evaluation point, not necessarily stationary).

	Posterior mode			Posterior draws			log MDD			
Prior on ρ	$\hat{\rho}$	(s.e.)	$\log L$	mean	sd	95% HPD	Laplace	CD − RE	bridge	Occam term
Beta$(0.85, 0.10)$ (reference)	0.897	(0.023)	4244.0	0.879	0.029	[0.820, 0.932]	4187.2	14.4	4190.6	-2.43
Beta$(0.95, 0.03)$	0.913	(0.029)	4244.5	–	–	–	4187.1	14.3	–	-2.60
Beta$(0.99, 0.01)$‡	0.934	(0.016)	4244.8	–	–	–	4181.5	8.7	–	-6.43
Uniform$[0, 1]$	0.895	(0.024)	4244.0	–	–	–	4185.8	13.0	–	-3.90
Beta$(0.70, 0.15)$	0.890	(0.024)	4243.7	–	–	–	4186.4	13.6	–	-3.30

Attention and shock volatility. Table OA.6 and Figure OA.4 give the diagnostics of §5.7.3. Besides the shock and measurement-error standard deviations, which together explain 27% of the posterior variance of ρ, all other parameters together explain 77%. The marginal standard error of ρ (0.023) is about three times its conditional one. The second-order rows of Panel B evaluate the curvature at the first-order mode.

Sector- and block-specific attention under Beta priors. Table OA.7 and Figure OA.5 estimate the sector-specific and block-specific restrictions of §5.5 with one Beta$(0.85, 0.10)$ prior per free attention parameter. Each reproduces the operator's likelihood to machine

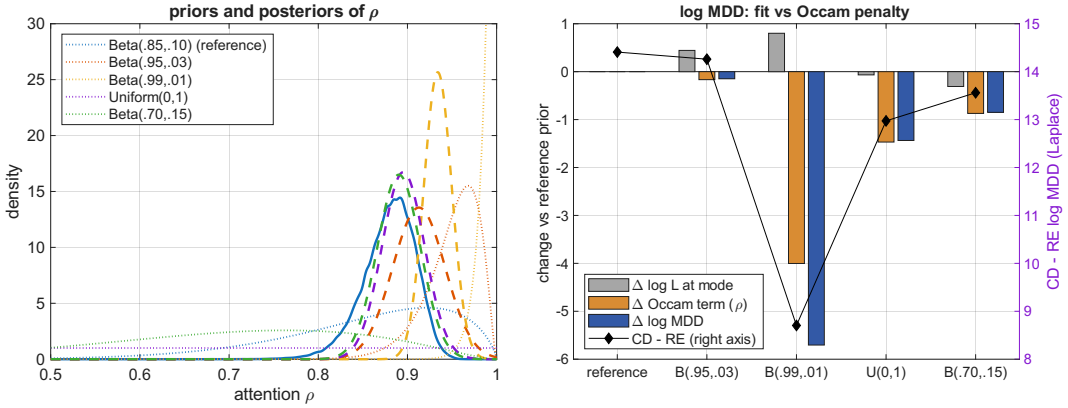


Figure OA.3: Common-attention restriction, prior sensitivity. Left: priors (dotted) and posteriors of ρ (solid: sampled, Beta$(0.85, 0.10)$ prior only; dashed: Laplace approximation at the mode). Right: change in the log marginal data density relative to the Beta$(0.85, 0.10)$ prior of §5.7 (labeled "reference" in the figure): the change in fit, the partial ρ-specific prior and curvature term ("Occam term"; it does not add up to the whole change), and the CD-over-RE margin.

Table OA.6: Common-attention restriction: what identifies attention. Panel A: posterior correlation of ρ with the shock and measurement-error parameters (all draws; draws in the main basin; implied by the Hessian at the mode). Panel B: two-dimensional log-likelihood slices in ρ and a preference-shock parameter at the mode, other parameters held fixed, at first and second order: Hessian-implied correlation and the standard error of ρ holding the second parameter fixed (cond.) or letting it adjust (marg.).

A. Posterior correlation with ρ (ranked by absolute value)

Parameter	all draws	main basin	Hessian	Parameter	all draws	main basin	Hessian
σ_{ζ_c}	+0.50	+0.50	+0.61	ρ_g	+0.03	+0.03	+0.05
σ_{π}^{me}	−0.20	−0.20	−0.19	$\sigma_{\Delta c}^{me}$	+0.03	+0.03	+0.06
σ_a	+0.11	+0.11	+0.11	σ_R^{me}	+0.02	+0.02	+0.00
ρ_{ζ_h}	+0.09	+0.22	+0.25	σ_{ζ_h}	+0.02	+0.03	+0.18
σ_{spr}^{me}	−0.07	−0.07	−0.01	ρ_θ	−0.02	−0.01	−0.00
ρ_{ζ_c}	+0.07	+0.07	+0.15	$\sigma_{\Delta w}^{me}$	−0.01	−0.02	−0.02
$\sigma_{\Delta y}^{me}$	−0.07	−0.07	−0.04	$\sigma_{\Delta g}^{me}$	−0.01	−0.01	−0.02
σ_θ	−0.05	−0.03	−0.00	ρ_{θ_w}	+0.00	+0.00	+0.00
σ_g	+0.05	+0.05	+0.05	σ_{θ_w}	−0.00	−0.01	+0.00

B. Log-likelihood slices at the mode

Parameter	Order	Hessian corr.	s.e.(ρ) cond.	s.e.(ρ) marg.	Ratio	Full Hessian: corr.; s.e.(ρ) fully marginal
σ_{ζ_c}	1	+0.22	0.0078	0.0080	1.03	+0.61; 0.0230
	2	+0.33	0.0057	0.0060	1.06	—
ρ_{ζ_c}	1	−0.41	0.0078	0.0086	1.09	+0.15; 0.0230
	2	−0.30	0.0057	0.0060	1.05	—

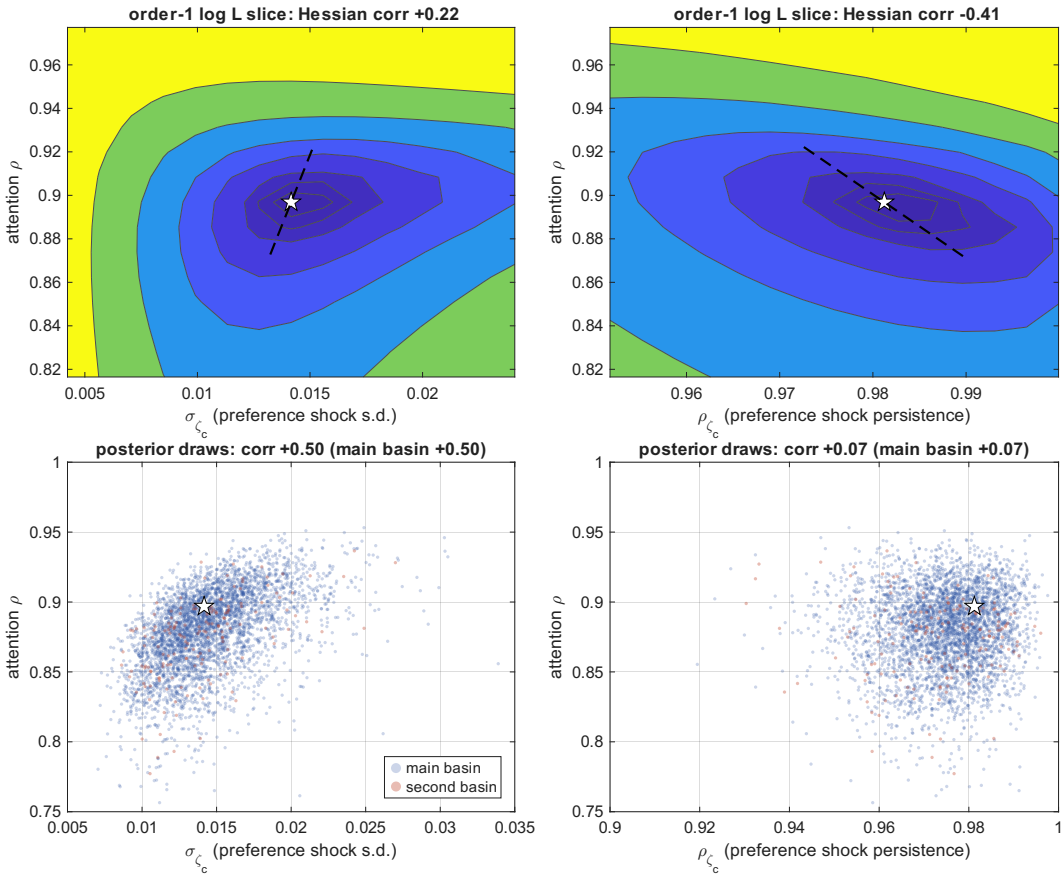


Figure OA.4: Common-attention restriction: attention and the preference shock, posterior draws and log-likelihood slices.

precision when all its discounts are equal. Against the single-attention restriction, the two-parameter sector-specific model raises the Laplace log marginal data density by 0.8 log points and the four-parameter block-specific model by 0.9. The household and Euler attentions inherit the identification of ρ (§5.7.3); the firm-side ones are nearly orthogonal to everything else. In a slice through the single-attention mode, with the other parameters held fixed, setting household attention to 0.99 costs about 140 log points of posterior, and setting firm attention to 0.99 about 2.

Table OA.7: Sector-specific (household/firm variable groups) and block-specific (per forward-looking condition) discounting under Beta priors, with rational expectations and the common-attention restriction. Posterior modes (global optimizer from dispersed starts); log marginal data density by Laplace; one Beta$(0.85, 0.10)$ prior per free attention parameter. Differences quoted in the text are computed from unrounded values.

Expectations	# attention parameters	log L	log MDD (Laplace)	Attention (mode, s.e.)
Rational expectations	0	4234.2	4172.8	—
Operator, common ρ	1	4244.0	4187.2	$\rho = 0.897$ (0.023)
Sector-specific (households vs. firms)	2	4244.6	4188.1	$\rho_H = 0.868$ (0.034), $\rho_F = 0.973$ (0.029)
Block-specific (one per forward condition)	4	4245.0	4188.2	$\rho_{\text{FOC}} = 0.972$ (0.032), $\rho_{\text{Euler}} = 0.854$ (0.039), $\rho_{\text{wage}} = 0.952$ (0.052), $\rho_{\text{price}} = 0.967$ (0.036)

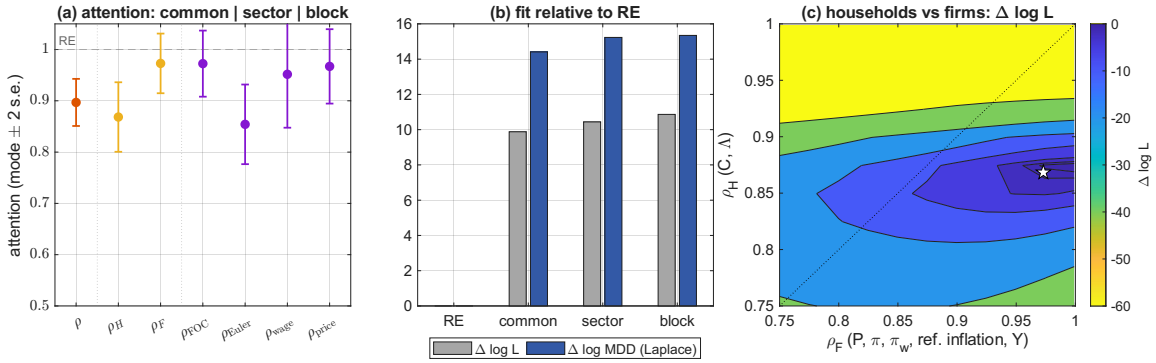


Figure OA.5: Sector-specific and block-specific discounting against the common-attention restriction: estimates and log-likelihood slices. Panel (c): log-likelihood relative to its maximum on the grid, other parameters at the households-vs-firms mode.

OA-F Lower-bound results

OA-F.1 The cost of the lower bound: full results

Table OA.8 reports the numbers behind Figure 8 of the main text (Section 4.5), for the occasionally binding lower bound.

Table OA.8: Cost of the lower bound by the number of quarters N it binds: size of the demand shock that produces the episode, cumulative output loss over 40 quarters per 100bp of policy space, the part of that loss due to the bound, and the amplification ratio (loss with the bound over loss without it). "n.m.": not meaningful, because at those parameters the demand shock that lowers the policy rate raises output. "restr.": the common attention of the restricted estimate (§5.7), other parameters at the calibration; "Posterior modes": the rational-expectations mode and the mode of the common-attention restriction.

quarters at the bound N	1	2	4	6	8	10	12
Calibration							
demand shock needed (s.d.)							
RE	288.5	291.9	304.4	321.3	342.0	369.1	411.4
CD, $\rho = 0.9$	73.3	75.0	80.0	86.5	97.7	110.6	129.1
CD, $\rho = 0.8$	54.6	55.9	59.1	65.0	74.0	81.8	98.3
CD, $\rho = 0.6$	53.6	54.3	57.9	63.2	70.5	82.8	96.4
CD, $\rho = 0.897$ (restr.)	72.1	73.8	78.7	85.2	96.2	108.8	127.3
cumulative output loss over 40 qtrs per 100bp of policy space (%-qtrs)							
RE	34.54	34.96	36.49	38.65	41.48	45.51	52.87
CD, $\rho = 0.9$	8.28	8.48	9.10	9.97	11.62	13.87	17.76
CD, $\rho = 0.8$	5.83	5.97	6.35	7.08	8.37	9.60	12.57
CD, $\rho = 0.6$	4.96	5.02	5.38	5.93	6.72	8.11	9.71
CD, $\rho = 0.897$ (restr.)	8.13	8.33	8.94	9.80	11.43	13.63	17.52
extra loss due to the bound, per 100bp of policy space (%-qtrs)							
RE	0.00	0.01	0.05	0.19	0.53	1.32	3.61
CD, $\rho = 0.9$	0.00	0.01	0.06	0.19	0.58	1.37	3.18
CD, $\rho = 0.8$	0.00	0.01	0.04	0.15	0.48	0.87	2.09
CD, $\rho = 0.6$	0.00	0.00	0.03	0.08	0.20	0.45	0.79
CD, $\rho = 0.897$ (restr.)	0.00	0.01	0.06	0.19	0.58	1.36	3.16
amplification by the bound (loss with / without)							
RE	1.000	1.000	1.001	1.005	1.013	1.030	1.073
CD, $\rho = 0.9$	1.000	1.001	1.007	1.019	1.053	1.110	1.218
CD, $\rho = 0.8$	1.000	1.001	1.007	1.022	1.060	1.100	1.199
CD, $\rho = 0.6$	1.000	1.000	1.005	1.014	1.030	1.059	1.088
CD, $\rho = 0.897$ (restr.)	1.000	1.001	1.007	1.020	1.053	1.110	1.220
Posterior modes (CD: common-attention restriction)							
demand shock needed (s.d.)							
RE (RE mode)	19.1	19.2	19.8	20.5	21.4	22.4	23.4
CD, $\hat{\rho} = 0.897$ (CD mode)	21.6	21.7	22.0	22.4	22.8	23.2	23.6
CD mode, $\rho = 1$ (RE)	8.8	8.8	9.0	9.2	9.5	9.7	10.0
CD mode, $\rho = 0.90$	22.9	23.0	23.3	23.7	24.1	24.5	24.9
CD mode, $\rho = 0.80$	11.5	11.5	11.7	11.8	12.0	12.3	12.5
CD mode, $\rho = 0.60$	11.6	11.6	11.7	11.9	12.1	12.4	12.7
cumulative output loss over 40 qtrs per 100bp of policy space (%-qtrs)							
RE (RE mode)	20.50	20.70	21.33	22.22	23.32	24.68	26.46
CD, $\hat{\rho} = 0.897$ (CD mode)	25.00	25.12	25.45	25.94	26.52	27.23	27.98
CD mode, $\rho = 1$ (RE)	n.m.: the shock that lowers R raises output						
CD mode, $\rho = 0.90$	26.27	26.39	26.74	27.25	27.86	28.59	29.37
CD mode, $\rho = 0.80$	14.75	14.80	15.02	15.26	15.60	16.08	16.54
CD mode, $\rho = 0.60$	14.69	14.73	14.91	15.09	15.44	15.78	16.25
extra loss due to the bound, per 100bp of policy space (%-qtrs)							
RE (RE mode)	0.00	0.00	0.03	0.11	0.28	0.63	1.26
CD, $\hat{\rho} = 0.897$ (CD mode)	0.00	0.00	0.02	0.06	0.17	0.38	0.71
CD mode, $\rho = 1$ (RE)	n.m.: the shock that lowers R raises output						
CD mode, $\rho = 0.90$	0.00	0.00	0.02	0.06	0.17	0.38	0.72
CD mode, $\rho = 0.80$	0.00	0.00	0.02	0.05	0.11	0.26	0.46
CD mode, $\rho = 0.60$	0.00	0.00	0.01	0.02	0.05	0.09	0.16
amplification by the bound (loss with / without)							
RE (RE mode)	1.000	1.000	1.001	1.005	1.012	1.026	1.050
CD, $\hat{\rho} = 0.897$ (CD mode)	1.000	1.000	1.001	1.002	1.006	1.014	1.026
CD mode, $\rho = 1$ (RE)	n.m.: the shock that lowers R raises output						
CD mode, $\rho = 0.90$	1.000	1.000	1.001	1.002	1.006	1.014	1.025
CD mode, $\rho = 0.80$	1.000	1.000	1.001	1.003	1.007	1.017	1.029
CD mode, $\rho = 0.60$	1.000	1.000	1.000	1.001	1.003	1.006	1.010

OA-F.2 Attention at the lower bound: the bound as a Markov regime

The main text models the lower bound as an occasionally binding constraint, with attention the same in and out of the bound. The operator also allows attention to depend on the state of the economy (Proposition 4). This subsection uses that capability. The bound is a two-state Markov chain in the style of Binning and Maih (2017): in the floor regime the policy rate is at the floor and the Taylor feedback is switched off, and attention takes the value ρ_{floor}; in the normal regime it takes the value ρ_{normal}. Both are common across the eight discounted variables. All results are at the calibration, at order 1.

Existence. Under rational expectations the solution algorithms used here find no solution of the switching model that is stable regime by regime: the floor regime is a peg, which on its own does not anchor expectations. They find a solution that is stable regime by regime only when floor attention is below a threshold (Table OA.9). This is numerical evidence under the regime-by-regime criterion, not a proof that no other stable solution exists. The criterion is regime-by-regime stability, not the mean-square stability of Foerster et al. (2016), which can admit a transiently unstable regime. The baseline solution algorithm finds such solutions up to $\rho_{\text{floor}} = 0.729$, an algorithmic cutoff; above it, it returns no solution. The economic threshold, where the floor forward eigenvalue reaches one, is about 0.732. A Newton-type solver finds solutions with a floor forward eigenvalue below one up to the grid point 0.732. Above that point, it finds a solution whose floor eigenvalue exceeds one. Linear extrapolation of the floor forward eigenvalue places the crossing at about 0.732. The baseline-algorithm cutoff does not depend on normal-times attention or on the expected duration of the floor regime (exit probabilities from 0.02 to 0.25). These values are close to the threshold for a determinate permanent peg in the occasionally binding model of the main text ($\rho^* = 0.736$; Section 4.1): the two ways of modeling the bound give similar attention thresholds for a well-behaved peg.

Table OA.9: Solvability of the switching model in floor attention ρ_{floor}.

ρ_{normal}	1 (RE)	0.9	0.8	0.6
largest solvable ρ_{floor}, baseline solution algorithm	0.729	0.729	0.729	0.729
Newton-type solver, $\rho_{\text{normal}} = 1$: solutions with $\max\lvert\lambda(\mathcal{F}_f)\rvert < 1$ up to $\rho_{\text{floor}} = 0.732$; at 0.740 it finds one with $\max\lvert\lambda(\mathcal{F}_f)\rvert = 1.019 > 1$				
floor forward eigenvalue $\max\lvert\lambda(\mathcal{F}_f)\rvert$ reaches one at $\rho_{\text{floor}} = 0.7320$ (linear extrapolation)				
$\max\lvert\lambda(\mathcal{F}_f)\rvert$: 0.9732 at 0.720, 0.9819 at 0.724, 0.9908 at 0.728				
baseline-algorithm threshold for floor exit probability $p_{fn} \in [0.02, 0.25]$ ($\rho_{\text{normal}} = 1$): 0.729				

Forward guidance. Below the threshold, the effect of calendar forward guidance at the floor is bounded, and its limit is set by floor attention alone (Table OA.10, Figure OA.6):

the limiting multiplier of a calendar peg is 15.6 at $\rho_{\text{floor}} = 0.7$, 4.7 at 0.6, and 3.0 at 0.5, whatever attention is in normal times.

Table OA.10: Forward-propagation eigenvalues in the normal and floor regimes, and the calendar-peg multiplier (output today, relative to a one-quarter peg) at $K = 20$, $K = 80$, and in the limit.

ρ_{floor}	ρ_{normal}	$\max \lvert\lambda(\mathcal{F}_n)\rvert$	$\max \lvert\lambda(\mathcal{F}_f)\rvert$	peg K=20	peg K=80	peg limit
0.7	1 (RE)	0.925	0.932	11.97	15.50	15.56
0.7	0.9	0.869	0.932	11.97	15.50	15.56
0.7	0.8	0.809	0.932	11.97	15.50	15.56
0.7	0.6	0.647	0.932	11.97	15.50	15.56
0.6	1 (RE)	0.925	0.761	4.68	4.70	4.70
0.6	0.9	0.869	0.761	4.68	4.70	4.70
0.6	0.8	0.809	0.761	4.68	4.70	4.70
0.6	0.6	0.647	0.761	4.68	4.70	4.70
0.5	1 (RE)	0.925	0.616	2.95	2.95	2.95
0.5	0.9	0.869	0.616	2.95	2.95	2.95
0.5	0.8	0.809	0.616	2.95	2.95	2.95
0.5	0.6	0.647	0.616	2.95	2.95	2.95

The cost of the lower bound. Here, unlike in the main text, the episode length N is imposed on the simulated regime path: a one-standard-deviation demand shock hits at the floor, and the floor regime is held for N quarters. Agents' decision rules are those of the Markov chain, so they do not know that the episode will last N quarters; the duration is imposed on the simulation, not announced. Floor attention shapes how the cost varies with duration (Table OA.11, Figure OA.7); unless stated otherwise the comparisons below hold normal times rational and compare the endpoints $N = 1$ and $N = 12$. With $\rho_{\text{floor}} = 0.7$ the loss rises over that range (+32%). With $\rho_{\text{floor}} = 0.6$ the endpoints are about equal (+0%), but the profile is not flat: it falls from 0.993 to 0.924 and returns to 0.998. With $\rho_{\text{floor}} = 0.5$ the loss falls (−13%). Several profiles are nonmonotone. Lower attention in normal times mainly reduces the loss of short episodes and makes the profile rise more steeply: with $\rho_{\text{normal}} = 0.6$ the change from $N = 1$ to $N = 12$ is +26% at $\rho_{\text{floor}} = 0.6$ and +8% at $\rho_{\text{floor}} = 0.5$. Its effect on long episodes is small and of either sign (at $\rho_{\text{floor}} = 0.5$ and $N = 12$ the loss is 0.851 with rational normal times and 0.848 with $\rho_{\text{normal}} = 0.6$).

Fiscal policy. The government-spending multiplier at the floor is about one on impact and barely depends on attention; its present value rises as normal-times attention falls, as in the main text (Table OA.12, Figure OA.8).

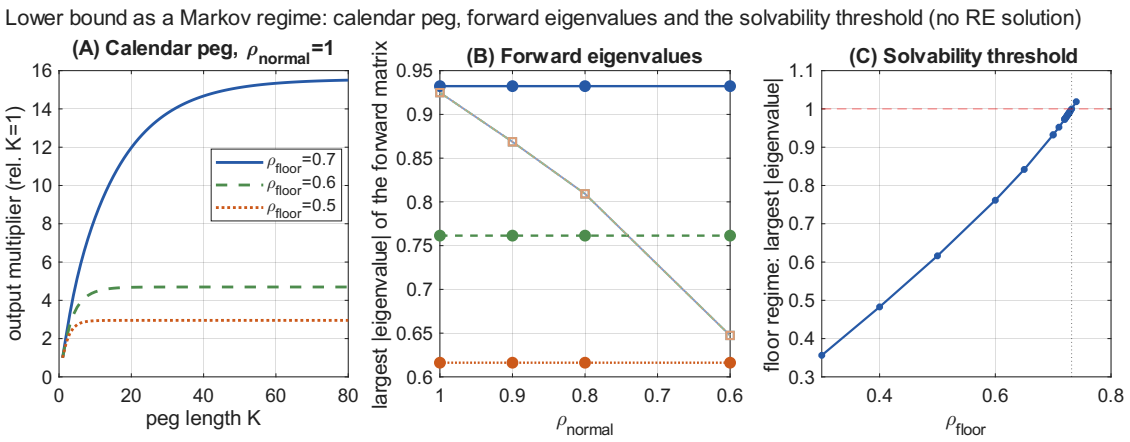


Figure OA.6: Calendar forward guidance at the floor, switching model, by floor and normal attention. Panel (B): circles, floor regime; squares, normal regime. Panel (C): the solution algorithms used find a solution while the largest floor-regime eigenvalue is below one; the dotted vertical line marks $\rho_f^* = 0.732$, where it reaches one. "No RE solution" in the title: none found under rational expectations by those algorithms, not a proof that none exists.

Table OA.11: Cumulative output loss over 40 quarters (percent-quarters) of a one-standard-deviation demand shock at the floor, with the floor held N quarters.

ρ_{floor}	ρ_{normal}	$N=1$	$N=2$	$N=4$	$N=6$	$N=8$	$N=10$	$N=12$	N=12 vs 1
0.7	1 (RE)	1.010	1.003	1.021	1.079	1.161	1.249	1.333	+32%
0.7	0.9	0.965	0.971	1.012	1.085	1.175	1.265	1.349	+40%
0.7	0.8	0.923	0.941	1.004	1.090	1.186	1.279	1.363	+48%
0.7	0.6	0.826	0.869	0.974	1.088	1.198	1.298	1.382	+67%
0.6	1 (RE)	0.993	0.964	0.929	0.924	0.941	0.968	0.998	+0%
0.6	0.9	0.945	0.927	0.909	0.917	0.941	0.971	1.002	+6%
0.6	0.8	0.901	0.892	0.891	0.910	0.940	0.974	1.006	+12%
0.6	0.6	0.799	0.811	0.845	0.887	0.931	0.972	1.007	+26%
0.5	1 (RE)	0.982	0.941	0.879	0.847	0.838	0.841	0.851	-13%
0.5	0.9	0.933	0.901	0.856	0.836	0.833	0.840	0.851	-9%
0.5	0.8	0.888	0.865	0.835	0.825	0.828	0.839	0.852	-4%
0.5	0.6	0.785	0.781	0.783	0.795	0.812	0.830	0.848	+8%

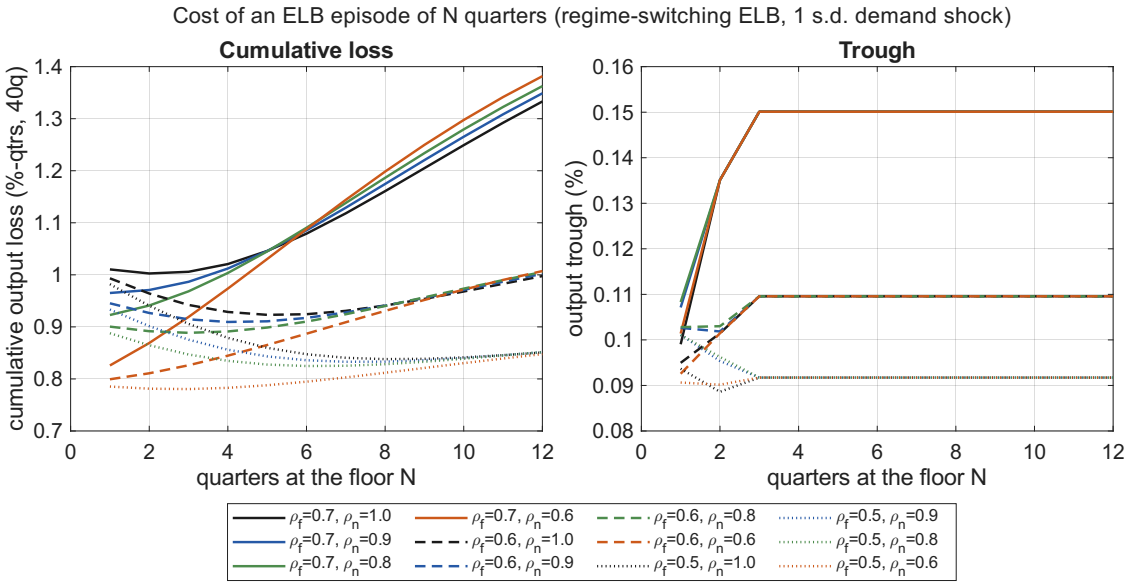


Figure OA.7: Cost of a lower-bound episode of imposed length N, switching model.

Table OA.12: Output multiplier dY/dG (elasticity $d\ln Y/d\ln G$ in brackets), normal regime and at the floor (eight quarters), impact and present value.

		normal regime		at the floor ($N=8$)	
ρ_{floor}	ρ_{normal}	impact	PV	impact	PV
0.7	1 (RE)	0.803 [0.161]	0.448 [0.090]	1.059 [0.212]	0.885 [0.177]
0.7	0.9	0.859 [0.172]	0.566 [0.113]	1.059 [0.212]	0.929 [0.186]
0.7	0.8	0.905 [0.181]	0.665 [0.133]	1.059 [0.212]	0.967 [0.193]
0.7	0.6	0.959 [0.192]	0.804 [0.161]	1.059 [0.212]	1.025 [0.205]
0.6	1 (RE)	0.803 [0.161]	0.448 [0.090]	1.018 [0.204]	0.823 [0.165]
0.6	0.9	0.859 [0.172]	0.566 [0.113]	1.018 [0.204]	0.862 [0.172]
0.6	0.8	0.905 [0.181]	0.665 [0.133]	1.018 [0.204]	0.896 [0.179]
0.6	0.6	0.959 [0.192]	0.804 [0.161]	1.018 [0.204]	0.948 [0.190]
0.5	1 (RE)	0.803 [0.161]	0.448 [0.090]	1.007 [0.201]	0.808 [0.162]
0.5	0.9	0.859 [0.172]	0.566 [0.113]	1.007 [0.201]	0.846 [0.169]
0.5	0.8	0.905 [0.181]	0.665 [0.133]	1.007 [0.201]	0.878 [0.176]
0.5	0.6	0.959 [0.192]	0.804 [0.161]	1.007 [0.201]	0.928 [0.186]

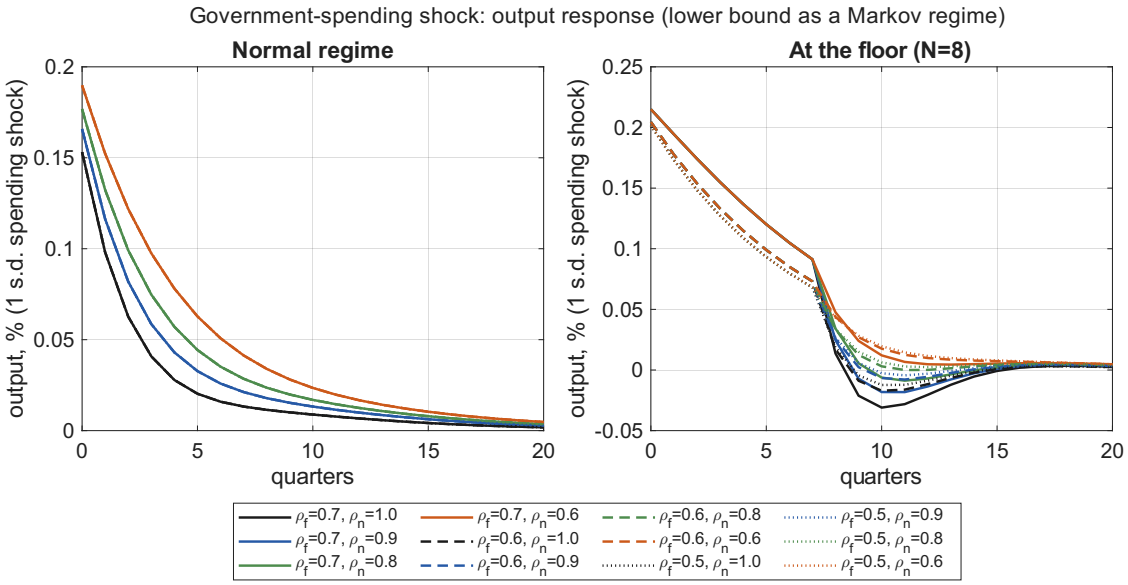


Figure OA.8: Government-spending multiplier, switching model.

Summary. Letting attention fall at the bound is a coherent way to obtain a well-behaved lower-bound regime: in this model, the solution algorithms used here then find a solution that is stable regime by regime. The price is a second, unestimated attention parameter, ρ_{floor}. The main text therefore keeps attention the same in and out of an occasionally binding bound, under which the same threshold is the condition for a determinate peg.

OA-G The laboratory model: derivations and variants

This section derives every equilibrium condition of the laboratory model, (A.1)–(A.31) in Appendix A, from the agents' problems and derives its balanced growth path. It then writes out the equilibrium conditions of each variant used in the paper. For the stationary policy variant, it writes out the private-sector conditions and the general form of the planner's conditions. The notation is that of Appendix A. The derivation is under rational expectations; the behavioral model replaces $\mathbb{E}_t$ by $\mathbb{E}_t^{\mathrm{BR}}$ in the resulting conditions (Definition 1), with the contractions of Appendix A.

OA-G.1 Agents

A continuum of households on $[0,1]$ each supplies a differentiated type of labor, consumes, saves in one-period nominal bonds, sets its wage subject to a Rotemberg cost, lends working capital to firms, pays lump-sum taxes and owns the firms. A labor packer aggregates the labor types into homogeneous labor. A continuum of monopolistically competitive firms on $[0,1]$ produces differentiated goods with labor, borrows the wage bill from households within the period, and sets prices subject to a Rotemberg cost. A final-good producer aggregates the goods. The government buys final goods and balances its budget with lump-sum taxes. The central bank sets the policy rate.

OA-G.2 Households

Preferences. Household j maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[\zeta_{C,t} \log(C_t - bC_{t-1}) - \zeta_{H,t} \frac{H_t(j)^{1+\nu}}{1+\nu} \Big], \tag{OA.13}$$

where the habit is internal: C_{t-1} is the household's own past consumption. The shifters $\zeta_{C,t}$ and $\zeta_{H,t}$ follow (A.1) and (A.2). Consumption and labor choices are the same for all households in equilibrium, so the index j is dropped where it plays no role.

Labor demand. The labor packer combines the labor types with a CES aggregator of elasticity $\theta_t^w > 1$, which follows (A.8). Cost minimization gives the demand for type j and the wage index,

$$H_t(j) = \Big(\frac{W_t(j)}{W_t} \Big)^{-\theta_t^w} H_t, \qquad W_t = \Big[\int_0^1 W_t(j)^{1-\theta_t^w} dj \Big]^{1/(1-\theta_t^w)}. \tag{OA.14}$$

Budget constraint. The household holds a nominal bond B_t. A bond bought at t pays $(1+R_t)(1+k_s s_t)$ at $t+1$: the credit spread s_t, which follows (A.24), is a wedge between the policy rate and the return on saving. Changing the wage costs $\frac{\phi_w}{2}\big(\frac{W_t(j)}{W_{t-1}(j)\tilde{\Pi}_t^w}-1\big)^2 P_t Y_t$ in nominal terms, where reference wage inflation $\tilde{\pi}_t^w$ follows (A.4). The household also lends the wage bill $W_t H_t$ to firms at the start of the period and receives $(1+k_s s_t)W_t H_t$ at its end, pays lump-sum taxes $-T_t$ and receives the firms' dividends $D_t - k_s s_t W_t H_t$. The nominal budget constraint is

$$P_t C_t + B_t + \frac{\phi_w}{2}\Big(\frac{W_t(j)}{W_{t-1}(j)\tilde{\Pi}_t^w}-1\Big)^2 P_t Y_t = (1+R_{t-1})(1+k_s s_{t-1})B_{t-1} + W_t(j)H_t(j) + T_t + D_t, \tag{OA.15}$$

in which the interest on working capital, $k_s s_t W_t H_t$, and the same amount deducted from the dividends cancel.

First-order conditions. Let $\beta^t \Lambda_t$ be the multiplier on (OA.15), so that Λ_t is the marginal utility of nominal wealth. The conditions for consumption and the bond are

$$\begin{aligned} C_t: &\quad \frac{\zeta_{C,t}}{C_t - bC_{t-1}} - \beta b\,\mathbb{E}_t \frac{\zeta_{C,t+1}}{C_{t+1}-bC_t} = \Lambda_t P_t, \\ B_t: &\quad \Lambda_t = \beta(1+R_t)(1+k_s s_t)\,\mathbb{E}_t \Lambda_{t+1}. \end{aligned}$$

The second term of the first condition is the effect of today's consumption on tomorrow's habit stock. These are (A.5) and (A.6) with $d=1$.

The household sets $W_t(j)$ taking (OA.14) into account, so that

$$\frac{\partial H_t(j)}{\partial W_t(j)} = -\theta_t^w \frac{H_t(j)}{W_t(j)}.$$

The terms of the Lagrangian that involve $W_t(j)$ are

$$\begin{aligned} -\zeta_{H,t}\frac{H_t(j)^{1+\nu}}{1+\nu} + \Lambda_t\Big[W_t(j)H_t(j) - \frac{\phi_w}{2}\Big(\frac{W_t(j)}{W_{t-1}(j)\tilde{\Pi}_t^w}-1\Big)^2 P_t Y_t\Big] \\ - \beta\,\mathbb{E}_t\Lambda_{t+1}\frac{\phi_w}{2}\Big(\frac{W_{t+1}(j)}{W_t(j)\tilde{\Pi}_{t+1}^w}-1\Big)^2 P_{t+1}Y_{t+1}. \end{aligned}$$

Differentiating with respect to $W_t(j)$ gives

$$\theta_t^w \zeta_{H,t} \frac{H_t(j)^{1+\nu}}{W_t(j)} + (1-\theta_t^w)\Lambda_t H_t(j) - \Lambda_t \phi_w \Big(\frac{W_t(j)}{W_{t-1}(j)\tilde{\Pi}_t^w} - 1 \Big) \frac{P_t Y_t}{W_{t-1}(j)\tilde{\Pi}_t^w} + \beta\, \mathbb{E}_t \Lambda_{t+1} \phi_w \Big(\frac{W_{t+1}(j)}{W_t(j)\tilde{\Pi}_{t+1}^w} - 1 \Big) \frac{W_{t+1}(j)}{W_t(j)^2 \tilde{\Pi}_{t+1}^w} P_{t+1} Y_{t+1} = 0.$$

In a symmetric equilibrium $W_t(j) = W_t$ and $H_t(j) = H_t$. Multiplying by W_t/H_t and using $\Pi_t^w = W_t/W_{t-1}$ gives

$$0 = \theta_t^w \zeta_{H,t} H_t^\nu - (\theta_t^w - 1)\Lambda_t W_t - \phi_w \frac{\Lambda_t P_t Y_t \Pi_t^w}{H_t \tilde{\Pi}_t^w} \Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1 \Big) + \beta\, \phi_w\, \mathbb{E}_t \frac{\Lambda_{t+1} P_{t+1} Y_{t+1} \Pi_{t+1}^w}{H_t \tilde{\Pi}_{t+1}^w} \Big(\frac{\Pi_{t+1}^w}{\tilde{\Pi}_{t+1}^w} - 1 \Big),$$

which is (A.7) with $d = 1$. The forward term carries $1/H_t$ and not $1/(H_t W_t)$, because the derivative of next period's wage growth with respect to $W_t(j)$ supplies a factor $1/W_t$; every term is then stationary along the balanced growth path. Without the adjustment cost the condition is $\theta_t^w \zeta_{H,t} H_t^\nu = (\theta_t^w - 1)\Lambda_t W_t$: the real wage is the markup $\theta_t^w/(\theta_t^w - 1)$ over the marginal rate of substitution $\zeta_{H,t} H_t^\nu/(\Lambda_t P_t)$.

The discount-factor shifter. The shifter d multiplies β in (A.5), (A.6), (A.7) and (A.13). It equals one except in the floor regime of the two-regime version, where its value d_{floor} makes the floor regime's steady state consistent with a policy rate at the bound (Binning and Maih, 2017). The shifter is imposed on these conditions; the household's problem above is written for $d = 1$.

OA-G.3 Final-good producer

A competitive producer combines the goods with a CES aggregator of elasticity $\theta_t > 1$, which follows (A.9). Profit maximization gives the demand for good i and the price index,

$$Y_t(i) = \Big(\frac{P_t(i)}{P_t} \Big)^{-\theta_t} Y_t, \qquad P_t = \Big[\int_0^1 P_t(i)^{1-\theta_t} di \Big]^{1/(1-\theta_t)}. \tag{OA.16}$$

OA-G.4 Firms

Technology and marginal cost. Firm i produces $Y_t(i) = A_t H_t(i)$, with technology following (A.10). It pays the wage bill before production, with funds borrowed from households

at the net rate $k_s s_t$ within the period, so the unit labor cost is $(1+k_s s_t)W_t$. The real marginal cost is therefore (A.11),

$$S_t = \frac{(1 + k_s s_t)W_t}{P_t A_t}.$$

Price setting. Changing the price costs, in nominal terms,

$$\frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{P_t(i)}{P_{t-1}(i)\tilde{\Pi}_t} - 1\Big)^2 P_t Y_t,$$

where reference price inflation $\tilde{\pi}_t$ follows (A.12). The firm is owned by households and values nominal profits with their marginal utility of nominal wealth. It chooses $P_t(i)$ to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Lambda_t \Big[P_t(i)Y_t(i) - (1 + k_s s_t)W_t \frac{Y_t(i)}{A_t} - \frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{P_t(i)}{P_{t-1}(i)\tilde{\Pi}_t} - 1\Big)^2 P_t Y_t\Big]$$

subject to (OA.16). With $\partial Y_t(i)/\partial P_t(i) = -\theta_t Y_t(i)/P_t(i)$, the first-order condition is

$$\Lambda_t\Big[(1-\theta_t)Y_t(i) + \theta_t \frac{(1+k_s s_t)W_t}{A_t}\frac{Y_t(i)}{P_t(i)} - \frac{\phi_p}{\tilde{\Pi}_t}\Big(\frac{P_t(i)}{P_{t-1}(i)\tilde{\Pi}_t} - 1\Big)\frac{P_t Y_t}{P_{t-1}(i)\tilde{\Pi}_t}\Big]$$
$$+ \beta\, \mathbb{E}_t \Lambda_{t+1} \frac{\phi_p}{\tilde{\Pi}_{t+1}}\Big(\frac{P_{t+1}(i)}{P_t(i)\tilde{\Pi}_{t+1}} - 1\Big)\frac{P_{t+1}(i)}{P_t(i)^2 \tilde{\Pi}_{t+1}} P_{t+1} Y_{t+1} = 0.$$

In a symmetric equilibrium $P_t(i) = P_t$. Dividing by $\Lambda_t Y_t$ and using $\Pi_t = P_t/P_{t-1}$ gives

$$0 = (1-\theta_t) + \theta_t S_t - \phi_p \frac{\Pi_t}{\tilde{\Pi}_t^2}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big) + \beta\, \phi_p\, \mathbb{E}_t \frac{\Lambda_{t+1} Y_{t+1} \Pi_{t+1}^2}{\Lambda_t Y_t \tilde{\Pi}_{t+1}^2}\Big(\frac{\Pi_{t+1}}{\tilde{\Pi}_{t+1}} - 1\Big),$$

which is (A.13) with $d = 1$ and $\chi_s = 0$. The term $\chi_s s_t \theta_t$ of (A.13) is a reduced-form channel through which the spread shifts prices directly, as an extra real cost $\chi_s s_t$ per unit of output would; that cost is not deducted from profits or resources. It is zero in every computation in the paper ($\chi_s = 0$).

Profits. Operating profits, before the financing cost of the wage bill, are (A.14),

$$D_t = P_t Y_t - W_t H_t - \frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big)^2 P_t Y_t.$$

The firm pays $D_t - k_s s_t W_t H_t$ to its owners; the financing cost goes to the households that lent the wage bill.

OA-G.5 Government and monetary policy

Government spending is the share g_t of output, (A.15)–(A.16), and the budget is balanced with lump-sum taxes: the transfer in (OA.15) is $T_t = -P_t G_t$. The central bank sets the shadow rate by the Taylor rule (A.17), and the policy rate is (A.18). The output gap (A.19) measures output relative to technology and steady-state hours, so it is zero on the balanced growth path.

OA-G.6 Market clearing and the resource constraint

The bond is in zero net supply, $B_t = 0$; this is imposed after the first-order conditions are derived. With $B_t = B_{t-1} = 0$, symmetry, $T_t = -P_t G_t$ and the profits above, the budget constraint (OA.15) becomes

$$P_t C_t + \frac{\phi_w}{2}\Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1\Big)^2 P_t Y_t = W_t H_t - P_t G_t + P_t Y_t - W_t H_t - \frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big)^2 P_t Y_t.$$

The wage bill cancels; dividing by P_t gives the resource constraint (A.3). Labor-market clearing and the symmetric production functions give $Y_t = A_t H_t$, (A.22). The remaining conditions are definitions: price and wage inflation, (A.20)–(A.21); output growth, (A.23); and the observation equations, which map the model into the seven series of Section 5. The system has 31 conditions for 31 endogenous variables; the budget constraint, the bond and the taxes are not carried, since they are implied by the rest.

OA-G.7 Balanced growth path

The balanced growth path is found by setting the shocks to zero, replacing X_{t+1} by $\Gamma_X X_t$ and X_{t-1} by X_t/Γ_X, where Γ_X is the gross growth rate of X (one for stationary variables), and solving.

Growth rates. From (A.10), $\Gamma_A = g_z \equiv 1 + \mu_a$. Hours are stationary, so (A.22) gives $\Gamma_Y = g_z$, and (A.3) and (A.16) give $\Gamma_C = \Gamma_G = g_z$. The Taylor rule gives $\pi_t = \pi^*$ (below), so $\Gamma_P = 1 + \pi^*$. Marginal cost is stationary, so (A.11) gives $\Gamma_W = (1+\pi^*)g_z$, and (A.14) gives $\Gamma_D = (1+\pi^*)g_z$. With logarithmic utility $\Lambda_t P_t \propto 1/C_t$ by (A.5), so $\Gamma_\Lambda = 1/[(1+\pi^*)g_z]$.

Exogenous processes and inflation. The autoregressive processes give $\zeta_C = \bar{\zeta}_C$, $\zeta_H = \bar{\zeta}_H$, $\theta = \bar{\theta}$, $\theta^w = \bar{\theta}^w$, $g = \bar{g}$ and $s = 0$. On the balanced growth path $R_t^* = R_{t-1}^* = R_{ss}^*$ and $\tilde{y}_t = 0$, so the Taylor rule (A.17) reduces to $(1-\rho_r)\phi_\pi[\log \Pi_t - \log(1+\pi^*)] = 0$:

inflation equals the target, $\pi = \pi^*$. Then (A.12) gives $\tilde{\pi} = \pi^*$; the wage identity (A.21) gives $1 + \pi^w = \Gamma_W = (1 + \pi^*)g_z$; and (A.4) gives $\tilde{\pi}^w = \pi^w$.

Marginal cost and the wage. With $\pi = \tilde{\pi}$ the adjustment-cost terms of (A.13) vanish, so $S = (\bar{\theta} - 1)/\bar{\theta}$, and (A.11) with $s = 0$ gives $W = SPA$.

Marginal utility and the interest rate. With $C_{t-1} = C_t/g_z$ and $C_{t+1} = g_z C_t$, (A.5) gives

$$\Lambda P = \frac{\zeta_C}{C}\Big[\frac{g_z}{g_z - b} - \frac{\beta d\, b}{g_z - b}\Big] = \frac{\zeta_C\,(g_z - \beta d\, b)}{(g_z - b)\,C}.$$

With $\Lambda_{t+1} = \Lambda_t/[(1 + \pi^*)g_z]$, the Euler equation (A.6) gives $1 + R = (1 + \pi^*)g_z/(\beta d)$. With $d = 1$ this is the shadow rate, $1 + R^*_{ss} = (1 + \pi^*)g_z/\beta$, and $R = R^*$ as long as $R^*_{ss} > \underline{r}$. In the floor regime of the two-regime version, $R = \underline{r}$ and $d = d_{\text{floor}} = (1 + \pi^*)g_z/[\beta(1 + \underline{r})]$, while the shadow rate keeps the value $(1 + \pi^*)g_z/\beta - 1$.

Hours. With $\pi^w = \tilde{\pi}^w$ the adjustment-cost terms of (A.7) vanish, so $\bar{\theta}^w \bar{\zeta}_H H^\nu = (\bar{\theta}^w - 1)\Lambda W$. The resource constraint and government spending give $C = (1 - \bar{g})Y = (1 - \bar{g})AH$. Multiplying ΛP by $W/P = SA$,

$$\Lambda W = \frac{\bar{\zeta}_C(g_z - \beta d\, b)(\bar{\theta} - 1)}{(g_z - b)(1 - \bar{g})\bar{\theta}\, H},$$

in which technology cancels because utility is logarithmic. Substituting,

$$H^{1+\nu} = \frac{(\bar{\theta}^w - 1)\,\bar{\zeta}_C\,(g_z - \beta d\, b)\,(\bar{\theta} - 1)}{(g_z - b)\,(1 - \bar{g})\,\bar{\theta}^w\,\bar{\theta}\,\bar{\zeta}_H}.$$

At the calibration ($d = 1$) this gives $H \approx 0.330$.

The rest. Then $Y = AH$, $C = (1 - \bar{g})Y$, $G = \bar{g}Y$, $\Lambda = \bar{\zeta}_C(g_z - \beta d\, b)/[(g_z - b)CP]$, $D = PY - WH = PY/\bar{\theta}$, $\tilde{y} = 0$ and $\Delta y = \log g_z$; the observables are $R^{obs} = R$ and zero for the other six. Technology and the price level are free normalizations, set to one at the base date. The solution is recursive: every value follows from the parameters and the values computed before it.

OA-G.8 The variants at a glance

Six variants of the laboratory model appear in the paper. Five of them replace only some conditions of Appendix A. Table OA.13 lists, for each of these five, the conditions it keeps and

the numbered conditions that replace the others. The sixth, the stationary policy variant, is a different model and is documented in §OA-G.13. As in Appendix A, every term dated $t+1$ is inside a conditional expectation. That expectation is $\mathbb{E}_t^{\mathrm{BR}}$ when the operator is applied. It is $\mathbb{E}_t$ in the block-specific and term-wise variants, whose discounting is written into the conditions.

The simplest variant is the version without the bound. It sets $d = 1$ and replaces (A.18) by

$$R_t = R_t^*. \tag{OA.17}$$

Table OA.13: Conditions of the model variants. Equation numbers without the prefix OA refer to Appendix A.

Variant	Conditions kept from Appendix A	Conditions replaced
No bound	(A.1)–(A.17), (A.19)–(A.31)	(A.18) by (OA.17)
Two-regime Markov bound (§OA-G.10)	(A.1)–(A.17), (A.19)–(A.31); d, σ_r, H_{ss}, the references (A.32) and the attentions are regime-specific	(A.18) by (A.18′)
Block-specific attention (§OA-G.11)	(A.1)–(A.4), (A.8)–(A.12), (A.14)–(A.31)	(A.5), (A.6), (A.7) and (A.13) by (OA.33)–(OA.36); the operator is not applied
Term-wise rule (§OA-G.12)	As for block-specific attention	(A.5), (A.6), (A.7) and (A.13) by (OA.37)–(OA.40); the operator is not applied
Detrended twin (§OA-G.9)	(A.1), (A.2), (A.4), (A.8), (A.9), (A.12), (A.15), (A.17), (A.18), (A.24), and the policy-rate, inflation and spread equations of (A.25)–(A.31); (A.3) and (A.16) in detrended variables	(A.5)–(A.7), (A.10), (A.11), (A.13), (A.14), (A.19), (A.21)–(A.23) and the four growth equations of (A.25)–(A.31) by (OA.18)–(OA.32); (A.20) is dropped

OA-G.9 The detrended twin

The levels model is checked against a stationary twin, written in detrended variables: $c_t = C_t/A_t$, $y_t = Y_t/A_t$, $\mathcal{G}_t = G_t/A_t$, $w_t = W_t/(P_tA_t)$, $\delta_t = D_t/(P_tA_t)$, $\lambda_t = \Lambda_t P_t A_t$, and technology growth $\gamma_t = A_t/A_{t-1}$. The price level is eliminated; inflation remains. Dividing the

conditions of Appendix A by their trends gives

$$\lambda_t = \frac{\zeta_{C,t}}{c_t - bc_{t-1}/\gamma_t} - \beta d\, b\, \frac{\zeta_{C,t+1}}{c_{t+1}\gamma_{t+1} - bc_t}, \tag{OA.18}$$

$$\lambda_t = \beta d\,(1+R_t)(1+k_s s_t)\,\frac{\lambda_{t+1}}{\Pi_{t+1}\gamma_{t+1}}, \tag{OA.19}$$

$$\begin{aligned} 0 = {} & \theta_t^w \zeta_{H,t} H_t^\nu - (\theta_t^w - 1)\lambda_t w_t - \phi_w \frac{\lambda_t y_t \Pi_t^w}{H_t \tilde{\Pi}_t^w}\Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1\Big) \\ & + \beta d\, \phi_w \frac{\lambda_{t+1} y_{t+1} \Pi_{t+1}^w}{H_t \tilde{\Pi}_{t+1}^w}\Big(\frac{\Pi_{t+1}^w}{\tilde{\Pi}_{t+1}^w} - 1\Big), \end{aligned} \tag{OA.20}$$

$$\log \gamma_t = \log(1+\mu_a) + \sigma_A \varepsilon_{A,t}, \tag{OA.21}$$

$$S_t = w_t(1 + k_s s_t), \tag{OA.22}$$

$$\begin{aligned} 0 = {} & (1-\theta_t) + \theta_t S_t + \chi_s s_t \theta_t - \phi_p \frac{\Pi_t}{\tilde{\Pi}_t^2}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big) \\ & + \beta d\, \phi_p \frac{\lambda_{t+1} y_{t+1} \Pi_{t+1}^2}{\lambda_t y_t \Pi_{t+1} \tilde{\Pi}_{t+1}^2}\Big(\frac{\Pi_{t+1}}{\tilde{\Pi}_{t+1}} - 1\Big), \end{aligned} \tag{OA.23}$$

$$\delta_t = y_t - w_t H_t - \frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big)^2 y_t, \tag{OA.24}$$

$$\tilde{y}_t = \log y_t - \log H_{ss}, \tag{OA.25}$$

$$\Pi_t^w = \frac{w_t}{w_{t-1}}\,\Pi_t\,\gamma_t, \tag{OA.26}$$

$$y_t = H_t, \tag{OA.27}$$

$$\Delta y_t = \log(y_t/y_{t-1}) + \log \gamma_t, \tag{OA.28}$$

$$\Delta y_t^{obs} = \log(y_t/y_{t-1}) + \log \gamma_t - \log(1+\mu_a) + \sigma_{me,y}\, e_t^y, \tag{OA.29}$$

$$\Delta c_t^{obs} = \log(c_t/c_{t-1}) + \log \gamma_t - \log(1+\mu_a) + \sigma_{me,c}\, e_t^c, \tag{OA.30}$$

$$\Delta w_t^{obs} = \log(w_t/w_{t-1}) + \log \gamma_t - \log(1+\mu_a) + \sigma_{me,w}\, e_t^w, \tag{OA.31}$$

$$\Delta g_t^{obs} = \log(\mathcal{G}_t/\mathcal{G}_{t-1}) + \log \gamma_t - \log(1+\mu_a) + \sigma_{me,g}\, e_t^g. \tag{OA.32}$$

These conditions replace their counterparts in Appendix A. The resource constraint (A.3) and government spending (A.16) keep their form, with Y, C and G replaced by y, c and $\mathcal{G}$. The price-inflation identity (A.20) drops out with the price level. The other conditions are unchanged (Table OA.13). At the base date of the balanced growth path ($A = P = 1$), the detrended values equal the levels, so the steady state of §OA-G.7 applies.

Under rational expectations, the twin is the levels model written in other variables. The two give the same likelihood to machine precision. This is the check reported in Section 3.

Under cognitive discounting, the operator acts on different objects in the two formulations. In the twin, it would contract $\{c, \lambda, y, \pi, \pi^w, \tilde{\pi}, \tilde{\pi}^w\}$ toward constant steady states. Net

inflation would be contracted in the level chart, and $\lambda = \Lambda PA$ would be contracted as one variable. The levels model instead contracts Λ and P separately. It contracts the price level through gross inflation in the log chart.

When both formulations use the same common attention, they agree at first order. Their nonlinear contractions generally differ. The numerical twin check is restricted to rational expectations, and the twin is not used under cognitive discounting.

OA-G.10 The two-regime Markov bound

The model of Online Appendix OA-F.2 treats the bound as a regime. A Markov chain $z_t \in \{1, 2\}$ (normal, floor) with constant transition probabilities p_{12} and p_{21} selects the regime. The bound equation (A.18) is replaced by

$$R_t = a_{z_t} R_t^* + (1 - a_{z_t})\big(\underline{r} + \sigma_{r,z_t}\varepsilon_{R,t}\big), \tag{A.18$'$}$$

with $(a, d, \sigma_r) = (1, 1, \sigma_{r,1})$ in the normal regime and $(0, d_{\text{floor}}, \sigma_{r,2})$ in the floor regime. All other conditions of Appendix A keep their form. The shifter d_{z_t} multiplies β in (A.5), (A.6), (A.7) and (A.13). The standard deviation σ_{r,z_t} replaces σ_r in the Taylor rule (A.17), which keeps running in both regimes. In the floor regime the Taylor feedback does not reach the policy rate, which stays at the floor up to a small shock. Attention is regime-specific. Every discounted variable has attention ρ_{normal} in the normal regime and ρ_{floor} in the floor regime, and $\mathbb{E}_t^{\text{BR}}$ uses the attention of the current regime.

The floor regime has its own steady state: $R = \underline{r}$, $d_{\text{floor}} = (1 + \pi^*)(1 + \mu_a)/[\beta(1 + \underline{r})]$, and the formulas of §OA-G.7 with $d = d_{\text{floor}}$. Hours, output, consumption, government spending, profits and Λ are therefore lower than in the normal regime. At the calibration, hours are 0.3247 instead of 0.3300. The wage, marginal cost, the inflation rates and the shadow rate are unchanged. In the floor regime, H_{ss} in (A.19) and the references (A.32) take the floor-regime values.

The computations use the calibration of Table 16, with $p_{12} = 0.025$ and $p_{21} = 0.040$ unless stated otherwise in Online Appendix OA-F.2. With the floor regime made identical to the normal regime, the model is the version without the bound. This requires $a = 1$, hence $d = 1$, and the same policy-shock standard deviation and the same attention in both regimes; the transition probabilities are unchanged. The stability results of §4.2 are computed with the model in that form.

OA-G.11 Block-specific attention

In the block-specific restriction of §5.5, each forward-looking condition e has one attention ρ_e, applied to all of its variables dated $t+1$: ρ_{foc} in (A.5), ρ_{Euler} in (A.6), ρ_{wpc} in (A.7) and ρ_{ppc} in (A.13). In condition e, each level $X \in \{C, \Lambda, P, Y\}$ dated $t+1$ is replaced by $\bar{X}_{t+1}(X_{t+1}/\bar{X}_{t+1})^{\rho_e}$, with the references (A.32). Each rate $x \in \{\pi, \pi^w, \tilde{\pi}, \tilde{\pi}^w\}$ dated $t+1$ is replaced by $x_{ss} + \rho_e(x_{t+1} - x_{ss})$. The exogenous $\zeta_{C,t+1}$ is not discounted. These contractions are written into the four conditions, and the operator is not applied. Written out, the four conditions are

$$\Lambda_t P_t = \frac{\zeta_{C,t}}{C_t - bC_{t-1}} - \beta d\, b\, \frac{\zeta_{C,t+1}}{\bar{C}_{t+1}(C_{t+1}/\bar{C}_{t+1})^{\rho_{\text{foc}}} - bC_t}, \tag{OA.33}$$

$$\Lambda_t = \beta d\,(1+R_t)(1+k_s s_t)\,\bar{\Lambda}_{t+1}(\Lambda_{t+1}/\bar{\Lambda}_{t+1})^{\rho_{\text{Euler}}}, \tag{OA.34}$$

the wage Phillips curve

$$\begin{aligned} 0 = {} & \theta_t^w \zeta_{H,t} H_t^\nu - (\theta_t^w - 1)\Lambda_t W_t - \phi_w \frac{\Lambda_t P_t Y_t \Pi_t^w}{H_t \tilde{\Pi}_t^w}\Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1\Big) \\ & + \beta d\, \phi_w \frac{\bar{\Lambda}_{t+1}(\Lambda_{t+1}/\bar{\Lambda}_{t+1})^{\rho_{\text{wpc}}}\, \bar{P}_{t+1}(P_{t+1}/\bar{P}_{t+1})^{\rho_{\text{wpc}}}\, \bar{Y}_{t+1}(Y_{t+1}/\bar{Y}_{t+1})^{\rho_{\text{wpc}}}}{H_t} \\ & \times \frac{1 + \pi_{ss}^w + \rho_{\text{wpc}}(\pi_{t+1}^w - \pi_{ss}^w)}{1 + \tilde{\pi}_{ss}^w + \rho_{\text{wpc}}(\tilde{\pi}_{t+1}^w - \tilde{\pi}_{ss}^w)}\Big(\frac{1 + \pi_{ss}^w + \rho_{\text{wpc}}(\pi_{t+1}^w - \pi_{ss}^w)}{1 + \tilde{\pi}_{ss}^w + \rho_{\text{wpc}}(\tilde{\pi}_{t+1}^w - \tilde{\pi}_{ss}^w)} - 1\Big), \end{aligned} \tag{OA.35}$$

and the price Phillips curve

$$\begin{aligned} 0 = {} & (1 - \theta_t) + \theta_t S_t + \chi_s s_t \theta_t - \phi_p \frac{\Pi_t}{\tilde{\Pi}_t^2}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big) \\ & + \beta d\, \phi_p \frac{\bar{\Lambda}_{t+1}(\Lambda_{t+1}/\bar{\Lambda}_{t+1})^{\rho_{\text{ppc}}}\, \bar{Y}_{t+1}(Y_{t+1}/\bar{Y}_{t+1})^{\rho_{\text{ppc}}}}{\Lambda_t Y_t} \\ & \times \frac{\big[1 + \pi_{ss} + \rho_{\text{ppc}}(\pi_{t+1} - \pi_{ss})\big]^2}{\big[1 + \tilde{\pi}_{ss} + \rho_{\text{ppc}}(\tilde{\pi}_{t+1} - \tilde{\pi}_{ss})\big]^2}\Big(\frac{1 + \pi_{ss} + \rho_{\text{ppc}}(\pi_{t+1} - \pi_{ss})}{1 + \tilde{\pi}_{ss} + \rho_{\text{ppc}}(\tilde{\pi}_{t+1} - \tilde{\pi}_{ss})} - 1\Big). \end{aligned} \tag{OA.36}$$

All other conditions of Appendix A are unchanged. With all four attentions equal to ρ, these conditions are those of the variable-specific operator with every $\rho_X = \rho$. The sector-specific and households/firms restrictions of §5.5 are equalities among the ρ_X of the variable-specific operator and need no separate equations.

OA-G.12 The term-wise rule

The term-wise rule of §5.7.4 switches the operator off. It discounts the expectation of each forward-looking *term* q, written in stationary form, toward its steady-state value: $\mathbb{E}_t^{\mathrm{BR}} q = \bar{q} + M(\mathbb{E}_t q - \bar{q})$. The terms carry the time-t normalizers that make them stationary (A_t, $A_t P_t$ or P_t, and C_t in the consumption condition). The other time-t factors ($1/A_t$, $1/(A_t P_t)$, $1/H_t$, $1/(\Lambda_t Y_t P_t)$) stay outside. Let $g_z \equiv 1 + \mu_a$, $\tilde{c} \equiv C_{ss}/A_{ss}$ and $\tilde{\lambda} \equiv \Lambda_{ss} A_{ss} P_{ss}$ (detrended steady-state values, $d = 1$). The four forward-looking conditions become

$$\Lambda_t P_t = \frac{\zeta_{C,t}}{C_t - bC_{t-1}} - \frac{\beta d\, b}{A_t}\, \zeta_{C,t+1}^{1-\iota_\zeta} \Big[\bar{q}_{5,t} + M\Big(\frac{\zeta_{C,t+1}^{\iota_\zeta} A_t}{C_{t+1} - bC_t} - \bar{q}_{5,t} \Big) \Big], \tag{OA.37}$$

$$\bar{q}_{5,t} = \frac{\bar{\zeta}_C^{\,\iota_\zeta}}{g_z \tilde{c} - b\big[(1-\iota_h)\tilde{c} + \iota_h C_t/A_t\big]},$$

$$\Lambda_t = \frac{\beta d\,(1+R_t)(1+k_s s_t)}{A_t P_t} \Big[\bar{q}_6 + M\big(A_t P_t \Lambda_{t+1} - \bar{q}_6 \big) \Big], \qquad \bar{q}_6 = \frac{\tilde{\lambda}}{g_z(1+\pi^*)}, \tag{OA.38}$$

the wage Phillips curve

$$\begin{aligned} 0 = {} & \theta_t^w \zeta_{H,t} H_t^\nu - (\theta_t^w - 1)\Lambda_t W_t - \phi_w \frac{\Lambda_t P_t Y_t \Pi_t^w}{H_t \tilde{\Pi}_t^w} \Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1 \Big) \\ & + \frac{\beta d}{H_t}\, M\, \phi_w \frac{\Lambda_{t+1} P_{t+1} Y_{t+1} \Pi_{t+1}^w}{\tilde{\Pi}_{t+1}^w} \Big(\frac{\Pi_{t+1}^w}{\tilde{\Pi}_{t+1}^w} - 1 \Big), \end{aligned} \tag{OA.39}$$

and the price Phillips curve

$$\begin{aligned} 0 = {} & (1-\theta_t) + \theta_t S_t + \chi_s s_t \theta_t - \phi_p \frac{\Pi_t}{\tilde{\Pi}_t^2} \Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1 \Big) \\ & + \frac{\beta d}{\Lambda_t Y_t P_t}\, M\, \phi_p \frac{P_t \Lambda_{t+1} Y_{t+1} \Pi_{t+1}^2}{\tilde{\Pi}_{t+1}^2} \Big(\frac{\Pi_{t+1}}{\tilde{\Pi}_{t+1}} - 1 \Big). \end{aligned} \tag{OA.40}$$

In the two Phillips curves, the steady-state values of the discounted terms, $\bar{q}_7$ and $\bar{q}_{13}$, are zero, because $\Pi^w = \tilde{\Pi}^w$ and $\Pi = \tilde{\Pi}$ on the balanced growth path. Each discounted term is therefore M times the term.

Two fixed switches set the treatment of the consumption condition, the only term with content that is not dated $t+1$. With $\iota_\zeta = 1$, the exogenous $\zeta_{C,t+1}$ is inside the discounted term, as in the literal rule. With $\iota_\zeta = 0$, it is outside, as under the operator. With $\iota_h = 0$, the term is anchored at the constant $\bar{q}_5$, so the time-t habit argument is discounted too, as in the literal rule. With $\iota_h = 1$, the term is anchored at the actual C_t. The "literal" version is $(\iota_\zeta, \iota_h) = (1, 0)$ and the "first-order-matched" version is $(0, 1)$. With $M = 1$ every version is the rational model. All other conditions of Appendix A are unchanged.

OA-G.13 The stationary policy variant

The optimal-policy exercises of §4.7 use a stationary variant. Technology is stationary, $\log A_t = \rho_a \log A_{t-1} + \sigma_A \varepsilon_{A,t}$; the price level is eliminated, so Λ_t is real marginal utility, W_t the real wage and D_t real profits; there is no lower bound and $d = 1$. The equilibrium conditions are

$$
\begin{aligned}
&\text{(A.1)},\ \text{(A.2)},\ \text{(A.3)},\ \text{(A.4)},\ \text{(A.8)},\ \text{(A.9)},\ \text{(A.12)},\\
&\text{(A.15)},\ \text{(A.16)},\ \text{(A.19)},\ \text{(A.22)},\ \text{(A.23)},\ \text{(A.24)},\\
&\Lambda_t = \frac{\zeta_{C,t}}{C_t - bC_{t-1}} - \beta b \frac{\zeta_{C,t+1}}{C_{t+1} - bC_t}, && \text{(OA.41)}\\
&\Lambda_t = \beta(1+R_t)(1+k_s s_t)\frac{\Lambda_{t+1}}{\Pi_{t+1}}, && \text{(OA.42)}\\
&0 = \theta_t^w \zeta_{H,t} H_t^\nu - (\theta_t^w - 1)\Lambda_t W_t - \phi_w \frac{\Lambda_t Y_t \Pi_t^w}{H_t \tilde{\Pi}_t^w}\Big(\frac{\Pi_t^w}{\tilde{\Pi}_t^w} - 1\Big)\\
&\qquad + \beta\phi_w \frac{\Lambda_{t+1} Y_{t+1} \Pi_{t+1}^w}{H_t \tilde{\Pi}_{t+1}^w}\Big(\frac{\Pi_{t+1}^w}{\tilde{\Pi}_{t+1}^w} - 1\Big), && \text{(OA.43)}\\
&S_t = \frac{W_t}{A_t}(1+k_s s_t), \qquad \Pi_t^w = \frac{W_t}{W_{t-1}}\Pi_t, && \text{(OA.44)}\\
&D_t = Y_t - W_t H_t - \frac{\phi_p}{2\tilde{\Pi}_t}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big)^2 Y_t, && \text{(OA.45)}\\
&0 = (1-\theta_t) + \theta_t S_t + \chi_s s_t \theta_t - \phi_p \frac{\Pi_t}{\tilde{\Pi}_t^2}\Big(\frac{\Pi_t}{\tilde{\Pi}_t} - 1\Big)\\
&\qquad + \beta\phi_p \frac{\Lambda_{t+1} Y_{t+1} \Pi_{t+1}}{\Lambda_t Y_t \tilde{\Pi}_{t+1}^2}\Big(\frac{\Pi_{t+1}}{\tilde{\Pi}_{t+1}} - 1\Big), && \text{(OA.46)}
\end{aligned}
$$

where reference wage inflation is indexed to its steady state $\tilde{\pi}_{ss}^w = \pi^*$. The policy rate R_t is the planner's instrument. The planner's conditions are formed as the private agents' conditions are: first-order conditions under rational expectations first, the operator second. Write the private conditions above as $\mathbb{E}_t f(X_{t+1}, X_t, X_{t-1}, \varepsilon_t) = 0$, with R_t in X_t, and the period utility as $U(X_t, X_{t-1}) = \zeta_{C,t}\log(C_t - bC_{t-1}) - \zeta_{H,t} H_t^{1+\nu}/(1+\nu)$. Under commitment the planner maximizes $\mathbb{E}_0 \sum_t \beta^t U(X_t, X_{t-1})$ subject to these conditions. With multipliers λ_t and the Lagrangian $\mathbb{E}_0 \sum_t \beta^t \big[U(X_t, X_{t-1}) + \lambda_t^\top f(X_{t+1}, X_t, X_{t-1}, \varepsilon_t)\big]$, the first-order conditions with respect to X_t are

$$
\nabla_1 U_t + \mathbb{E}_t\big[\beta \nabla_2 U_{t+1} + f_{2,t}^\top \lambda_t + \beta f_{3,t+1}^\top \lambda_{t+1}\big] + \beta^{-1} f_{1,t-1}^\top \lambda_{t-1} = 0, \qquad \text{(OA.47)}
$$

where $\nabla_j U_s$ is the gradient of U with respect to its j-th argument at (X_s, X_{s-1}), and $f_{j,s}$ is the Jacobian of f with respect to its j-th argument, evaluated at $(X_{s+1}, X_s, X_{s-1}, \varepsilon_s)$. Under

discretion the planner solves the same problem without commitment, and the Markov-perfect rule is computed by the same perturbation method. The operator is then applied to the whole optimality system, the planner's conditions included, exactly as to the private conditions; the multipliers are not contracted. The planner therefore forms expectations as the private agents do. Equation (OA.47) is generated symbolically for the system above and is not expanded here. The operator contracts $\{C, \Lambda, Y, \pi, \pi^w, \tilde{\pi}, \tilde{\pi}^w\}$ with a common attention ρ toward their constant steady state, C, Λ and Y in the log chart and the rates in the level chart. Period welfare $W_t^{\text{flow}} = \zeta_{C,t} \log(C_t - bC_{t-1}) - \zeta_{H,t} H_t^{1+\nu}/(1+\nu)$ is recorded alongside and does not affect the equilibrium. The steady state is constant: $A = 1$, $\pi = \tilde{\pi} = \pi^w = \tilde{\pi}^w = \pi^*$, $S = (\bar{\theta} - 1)/\bar{\theta}$, $W = SA$, $1 + R = (1 + \pi^*)/\beta$,

$$H = \left[\frac{(\bar{\theta}^w - 1)\, \bar{\zeta}_C\, (1 - \beta b)\, (\bar{\theta} - 1)}{(1 - b)\, (1 - \bar{g})\, \bar{\theta}^w\, \bar{\theta}\, \bar{\zeta}_H} \right]^{1/(1+\nu)},$$

$Y = AH$, $C = (1 - \bar{g})Y$, $G = \bar{g}Y$, $\Lambda = \bar{\zeta}_C(1 - \beta b)/[(1 - b)C]$, $D = Y - WH$, $\tilde{y} = 0$ and $\Delta y = 0$. Technology persistence is $\rho_a = 0.90$; the other parameters are those of Table 16 or, where §4.7 says so, the estimated values.

References

Binning, A., and J. Maih (2017). "Modelling Occasionally Binding Constraints Using Regime-Switching." Norges Bank Working Paper 23/2017.

Foerster, A., J.F. Rubio-Ramírez, D.F. Waggoner, and T. Zha (2016). "Perturbation Methods for Markov-Switching Dynamic Stochastic General Equilibrium Models." *Quantitative Economics* 7(2), 637–669.

Gabaix, X. (2014). "A Sparsity-Based Model of Bounded Rationality." *Quarterly Journal of Economics* 129(4), 1661–1710.

Gabaix, X. (2016a). "Behavioral Macroeconomics via Sparse Dynamic Programming." NBER Working Paper 21848.

Gabaix, X. (2020). "A Behavioral New Keynesian Model." *American Economic Review* 110(8), 2271–2327.

McKay, A., E. Nakamura, and J. Steinsson (2016). "The Power of Forward Guidance Revisited." *American Economic Review* 106(10), 3133–3158.

Vehtari, A., A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner (2021). "Rank-Normalization, Folding, and Localization: An Improved $\widehat{R}$ for Assessing Convergence of MCMC." *Bayesian Analysis* 16(2), 667–718.